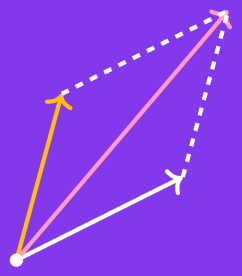
Linear Algebra, But Make It Make Sense

Lecture 1 – Vector spaces, subspaces, linear maps, rank & kernel



Sasha Sutton – personal study notes, Fall 2026

My notes + recap-with-proofs + fully solved exercises, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

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1 Survival kit: the notation you need before anything

Before we start, let's decode the symbols. Maths notation is basically a compressed language; once you can read it, half the "difficulty" disappears.

Symbol	How you read it	Example
$x \in E$	"x belongs to E" / "x is an element of E"	$3 \in \mathbb{R}$, $(1,2) \in \mathbb{R}^2$
$A \subseteq E$	"A is a subset of E" (everything in A is also in E)	$\mathbb{N} \subseteq \mathbb{R}$
$\forall$	"for all" / "for every"	$\forall x \in \mathbb{R},\ x^2 \geq 0$
$\exists$	"there exists (at least one)"	$\exists x \in \mathbb{R},\ x^2 = 4$
$\Rightarrow$	"implies" (if left is true, right is true)	$x = 2 \Rightarrow x^2 = 4$
$\iff$ / iff	"if and only if" (both directions)	$x^3 = 8 \iff x = 2$
$\{x \in E \mid P(x)\}$	"the set of x in E such that $P(x)$ holds"	$\{x \in \mathbb{R} \mid x > 0\}$
$f : X \to Y$	"f goes from X (domain) to Y (codomain)"	$f : \mathbb{R} \to \mathbb{R}$
$x \mapsto f(x)$	"x is sent to $f(x)$"	$x \mapsto x^2$
$\sum_{i=1}^{n} a_i$	$a_1 + a_2 + \cdots + a_n$ (always say the range of i!)	$\sum_{i=1}^{3} i = 6$
$E \times F$	Cartesian product: all pairs (x, y), $x \in E$, $y \in F$	$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$
$\cap$, $\cup$	intersection (in both), union (in at least one)	
$\mathbb{F}$	a field; in these notes always $\mathbb{F} = \mathbb{R}$	

COMMON L (don't do this) // Undefined notation

Every letter you write must be *introduced* ("Let $x \in \mathbb{R}^d \ldots$", "Let $\lambda \in \mathbb{R} \ldots$"). When you write a sum, **always** write the range $\sum_{i=\min}^{\max}$. A proof with mystery variables is an automatic ick for the grader.

The 4 proof "shapes" you'll use all the time

Most proofs follow one of these templates. Learn to recognise them:

1. **Existence and uniqueness.** First show *at least one* object exists, then assume there are two and show they're equal.
2. **Verify the properties.** "Show S is a subspace" $\Rightarrow$ check the 3 conditions one by one. Pure checklist energy.
3. **Induction (recurrence).** Prove for $n = 2$ (initialisation), then show "true for n" $\Rightarrow$ "true for $n+1$" (heredity).
4. **Equivalence.** To prove $A \iff B$, prove $A \Rightarrow B$, *then* $B \Rightarrow A$. Two separate mini-proofs.

2 Vector spaces: the arena where linear algebra happens

TL;DR // Chapter 2 in one breath

A vector space is a set where you can **add** things and **scale** things, and all the "normal" rules of arithmetic hold. Vectors don't have to be arrows: they can be lists of numbers, images, polynomials, functions, matrices... If you can add them and scale them nicely, they're vectors.

2.1 Warm-up: groups

DEFINITION // Group

Let G be a non-empty set with an operation $\star$. G is a **group** if:

(i) **Closure.** For all $x, y \in G$, $x \star y \in G$. *(you never leave the club)*

(ii) **Associativity.** For all $x, y, z \in G$, $(x \star y) \star z = x \star (y \star z)$. *(brackets don't matter)*

(iii) **Neutral element.** There is $e \in G$ with $x \star e = e \star x = x$ for all $x \in G$. *(the "do nothing" element)*

(iv) **Inverse.** For each $x \in G$ there is $x' \in G$ with $x \star x' = x' \star x = e$. *(every move can be undone)*

If moreover $x \star y = y \star x$ for all x, y, the group is **commutative** (a.k.a. **abelian**).

EXAMPLE // Groups and non-groups

- $(\mathbb{Z}, +)$: closure ✓, associativity ✓, neutral 0 ✓, inverse of n is $-n$ ✓. Commutative group. W.
- $(\mathbb{N}, +)$ with $\mathbb{N} = \{0, 1, 2, \dots\}$: the inverse of 3 would be $-3 \notin \mathbb{N}$. **Not a group.** L.
- $(\mathbb{R}, \times)$: 0 has no inverse ($0 \times x' = 1$ is impossible). Not a group. But $(\mathbb{R} \setminus \{0\}, \times)$ is one.

2.2 The definition of a vector space

DEFINITION // Vector space over $\mathbb{R}$

A **vector space** $(V, +, \cdot)$ over $\mathbb{R}$ is a set V with two operations:

$$+ : V \times V \to V,\ (x, y) \mapsto x + y \qquad \text{and} \qquad \cdot : \mathbb{R} \times V \to V,\ (a, x) \mapsto a \cdot x$$

such that:

(A) $(V, +)$ is a commutative group:

A1. Associativity: $(x + y) + z = x + (y + z)$ for all $x, y, z \in V$.

A2. Additive identity: there is $0 \in V$ with $x + 0 = x$ for all $x \in V$.

A3. Additive inverse: for every $x \in V$ there is $-x \in V$ with $x + (-x) = 0$.

A4. Commutativity: $x + y = y + x$ for all $x, y \in V$.

(B) the scalar multiplication behaves:

B1. Distributivity over scalar addition: $(\alpha + \beta)x = \alpha x + \beta x$.

B2. Distributivity over vector addition: $\alpha(x + y) = \alpha x + \alpha y$.

B3. "Mixed" associativity: $\alpha(\beta x) = (\alpha\beta)x$.

B4. Multiplicative identity: $1 \cdot x = x$.

Elements of V are called **vectors**; elements of $\mathbb{R}$ are called **scalars**.

Vector spaces can be defined over any **field** $\mathbb{F}$ (a structure closed under $+, -, \times, \div$, like $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$). In these notes, $\mathbb{F} = \mathbb{R}$.

COMMON L (don't do this) // "Can we subtract vectors?"

There is no separate "minus" operation in the axioms. But subtraction is just *adding the opposite*: $x - y := x + (-y)$. So yes, you can subtract. You *cannot* multiply two vectors or divide by a vector, though: that's not part of the deal.

2.3 The vector space hall of fame

EXAMPLE // Vector spaces you must know

1. $(\mathbb{R}, +, \cdot)$: real numbers are vectors (1D arrows on a line).
2. $\mathbb{R}$ is also a vector space *over* $\mathbb{Q}$; $\mathbb{C}$ is a vector space over $\mathbb{R}$ and over $\mathbb{Q}$. The field matters!
3. $(\mathbb{R}^n, +, \cdot)$, the **Euclidean space**. The main character of these notes.
4. $\mathcal{M}_{n,p}(\mathbb{R})$: all $n \times p$ real matrices (add entrywise, scale entrywise).
5. The set of all functions $I \to \mathbb{R}$ ($I \subseteq \mathbb{R}$ an interval): $(f+g)(t) = f(t)+g(t)$, $(\lambda f)(t) = \lambda f(t)$.
6. The set of real sequences $(u_n)_{n\in\mathbb{N}}$.
7. $\mathbb{R}_n[X] = \left\{\sum_{i=0}^{n} a_i X^i \;\middle|\; a_0, \dots, a_n \in \mathbb{R}\right\}$, polynomials of degree $\leq n$, with

$$(P+Q)(X) = \sum_{i=0}^{n} (a_i + b_i) X^i, \qquad (\lambda P)(X) = \sum_{i=0}^{n} \lambda a_i X^i.$$

2.4 Operations in $\mathbb{R}^n$ and what they look like

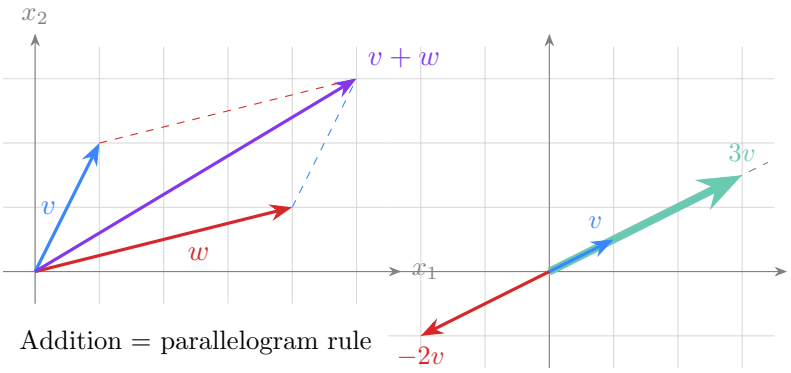
In $\mathbb{R}^n$ everything is done **component-wise**:

$$x + y = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \qquad \alpha x = \begin{pmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{pmatrix}.$$

EXAMPLE // Numbers, not vibes

Let $v = (1, 2)$ and $w = (3, -1)$ in $\mathbb{R}^2$.

1. $v + w = (1 + 3,\ 2 + (-1)) = (4, 1)$.
2. $2w = (6, -2)$, so $v + 2w = (7, 0)$.
3. $v - w = v + (-1)w = (-2, 3)$.



Addition = parallelogram rule

Scaling keeps the line, negative scalar flips the orientation

DEFINITION // Parallel vectors

Two vectors $v, w \in \mathbb{R}^n$ are **parallel** if there is a *non-zero* scalar $r \in \mathbb{R}$ with $w = rv$. Geometrically: same line through the origin. In $\mathbb{R}^n$ a vector has a **direction** and a **length** (norm); scaling changes the length, keeps the line, and a negative scalar reverses the orientation.

2.5 Product of vector spaces (stacking spaces like Lego)

DEFINITION // Cartesian product and product space

$A \times B = \{(a, b) \mid a \in A,\ b \in B\}$. If E, F are $\mathbb{F}$-vector spaces, define on $E \times F$:

$$(x, y) + (x', y') = (x + x',\ y + y'), \qquad \lambda \cdot (x, y) = (\lambda x,\ \lambda y).$$

THEOREM / PROPOSITION // Product space

With these operations, $(E \times F, +, \cdot)$ is an $\mathbb{F}$-vector space, the **product vector space**. Moreover, if $E_1, \ldots, E_n$ $(n > 1)$ are finite-dimensional,

$$\dim(E_1 \times E_2 \times \cdots \times E_n) = \sum_{i=1}^{n} \dim(E_i).$$

RECEIPTS (proof) // $E \times F$ is a vector space

Closure. Let $(x, y), (x', y') \in E \times F$. As E is a vector space, $x + x' \in E$; as F is, $y + y' \in F$. So $(x + x', y + y') \in E \times F$. Similarly, for $\lambda \in \mathbb{F}$, $\lambda x \in E$ and $\lambda y \in F$, so $\lambda(x, y) \in E \times F$.
Zero and opposites. The zero is $(0_E, 0_F)$ because $(x, y) + (0_E, 0_F) = (x + 0_E, y + 0_F) = (x, y)$. The opposite of (x, y) is $(-x, -y)$ since $(x, y) + (-x, -y) = (0_E, 0_F)$.
Other axioms. They all hold *component-wise* because they hold in E and in F. For instance $\lambda\big((x, y) + (x', y')\big) = (\lambda x + \lambda x', \lambda y + \lambda y') = \lambda(x, y) + \lambda(x', y')$. ■ *(and that's on periodt)*

RECEIPTS (proof) // Dimension of a product (induction)

Case $n = 2$. Let $(e_1, \ldots, e_p)$ be a basis of E and $(f_1, \ldots, f_q)$ a basis of F. Claim:

$$\mathcal{B} = \big((e_1, 0_F), \ldots, (e_p, 0_F), (0_E, f_1), \ldots, (0_E, f_q)\big)$$

is a basis of $E \times F$.
Spanning: for $(x, y) \in E \times F$ write $x = \sum_{i=1}^{p} \alpha_i e_i$, $y = \sum_{j=1}^{q} \beta_j f_j$. Then $(x, y) = \sum_{i=1}^{p} \alpha_i (e_i, 0_F) + \sum_{j=1}^{q} \beta_j (0_E, f_j)$.

Independence: if $\sum_{i=1}^{p} \alpha_i(e_i, 0_F) + \sum_{j=1}^{q} \beta_j(0_E, f_j) = (0_E, 0_F)$, then $\sum_i \alpha_i e_i = 0_E$ and $\sum_j \beta_j f_j = 0_F$; both families are bases so all $\alpha_i = 0$ and all $\beta_j = 0$.
So $\dim(E \times F) = p + q$.
Heredity. Assume the formula holds for some $n \geq 2$. Set $E' = E_1 \times \cdots \times E_n$, a vector space by the above. Using the case $n = 2$: $\dim(E' \times E_{n+1}) = \dim(E') + \dim(E_{n+1}) = \sum_{i=1}^{n+1} \dim(E_i)$.
By induction the formula holds for all $n > 1$. ■ *(and that's on periodt)*

EXAMPLE // $\mathbb{R}^n$ is a product

$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}}$, so $\dim \mathbb{R}^n = 1 + 1 + \cdots + 1 = n$. A vector $x = (x_1, \ldots, x_n)$ has **components** (entries) x_i. You can think of it as an $n \times 1$ **column vector**.

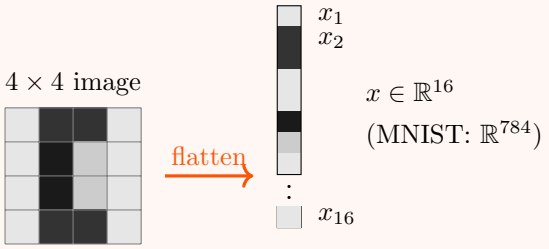
2.6 ML side quests: data *is* vectors

ML SIDE QUEST // Data points as vectors

In ML, a data point is a vector $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$ where each x_j is a **feature** (age, income, pixel intensity, word count…). A dataset with m observations is a collection $x^{(1)}, \ldots, x^{(m)} \in \mathbb{R}^d$.

ML SIDE QUEST // Images are vectors (MNIST)

A 28×28 grayscale image has 784 pixels. Read the pixels row by row and you get a vector in $\mathbb{R}^{784}$. MNIST: 70 000 images of digits 0–9, intensities in $[0, 1]$. Labels are **one-hot** vectors $y \in \mathbb{R}^{10}$. Classification = learning $f : \mathbb{R}^{784} \to \{0, \ldots, 9\}$ (in practice, f outputs a probability vector over 10 classes and we pick the max).



ML SIDE QUEST // Mean image, template matching, centering

Because images live in a vector space, you can **average** them. The mean image of class c is

$$\bar{x}_c = \frac{1}{m_c} \sum_{i=1}^{m_c} x^{(i)} \in \mathbb{R}^{784},$$

where the sum runs over the m_c training images of class c. This uses *only* addition and scaling (by $1/m_c$): pure vector-space operations.
Template matching classifier. For a new image x, compute the correlation with each class template:

$$\operatorname{corr}(\bar{x}_c, x) = \frac{\sum_{j=1}^{784} (\bar{x}_{c,j} - \bar{\bar{x}})(x_j - \bar{x})}{\sqrt{\sum_{j=1}^{784} (\bar{x}_{c,j} - \bar{\bar{x}})^2} \sqrt{\sum_{j=1}^{784} (x_j - \bar{x})^2}}, \qquad \hat{y} = \arg \max_{c \in \{0,\ldots,9\}} \operatorname{corr}(\bar{x}_c, x),$$

where $\bar{\bar{x}}$ is the mean pixel value of the template and $\bar{x}$ the mean pixel value of x.
Centering. Empirical mean $\bar{x} = \frac{1}{m} \sum_{i=1}^{m} x^{(i)}$, then $x^{(i)} \mapsto x^{(i)} - \bar{x}$. Basic preprocessing before PCA and many other algorithms.

ML SIDE QUEST // Multimodal data = product spaces

An object described by image features $u \in \mathbb{R}^{d_1}$ and text features $v \in \mathbb{R}^{d_2}$ lives in $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. In code you **concatenate**: $(u, v) \longleftrightarrow \begin{pmatrix} u \\ v \end{pmatrix} \in \mathbb{R}^{d_1+d_2}$. The dimension formula says $\dim = d_1 + d_2$. (Part 2 of the exercises is exactly this with $\mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3$.)

3 Subspaces, span, independence, bases, dimension

TL;DR // Chapter 3 in one breath

A **subspace** is a vector space living inside a bigger one: think of lines and planes *through the origin*. A **span** is everything you can build from some vectors. **Independent** means no vector is a copy-paste of the others. A **basis** is a family that is both spanning and independent: the minimal complete "alphabet". The number of vectors in a basis is the **dimension**.

3.1 Subspaces

DEFINITION // Subspace

Let V be a vector space. $S \subseteq V$ is a **subspace** of V if

(i) $0 \in S$;

(ii) S is closed under addition: $x, y \in S \Rightarrow x + y \in S$;

(iii) S is closed under scalar multiplication: $x \in S,\ \alpha \in \mathbb{R} \Rightarrow \alpha x \in S$.

French: *sous-espace vectoriel* (s.e.v.).

EXAM CHEAT CODE // "Show that S is a subspace of V"

1. Check $S \subseteq V$ (usually obvious, say it in one line).
2. **Zero:** plug 0 into the definition of S and check the condition holds.
3. **Stability:** take *arbitrary* $x, y \in S$ and $\lambda \in \mathbb{R}$. Write down what "$x \in S$" and "$y \in S$" mean (the equations they satisfy).
4. Show $x + y$ satisfies the equations, and λx satisfies them. (Speed-run: show $\lambda x + y \in S$ in one go; it covers both.)
5. Conclude: "Hence S is a subspace of V."

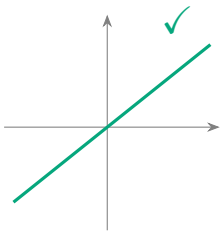
Faster alternatives (when allowed): S is a span; S is a kernel of a linear map; S is an intersection of subspaces; S is the solution set of homogeneous linear equations.

EXAM CHEAT CODE // "Show that S is *not* a subspace"

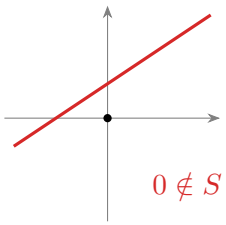
Find **one** explicit counterexample. Priority order:

1. Is $0 \notin S$? Done, instantly. (Any equation with a non-zero constant, e.g. $x_1 + x_2 = 1$.)
2. Otherwise find $x \in S$ and λ with $\lambda x \notin S$ (try $\lambda = -1$).
3. Otherwise find $x, y \in S$ with $x + y \notin S$.

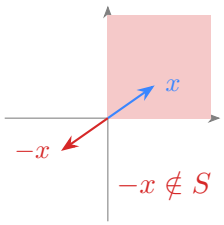
Use actual numbers. "It doesn't look like a subspace" is not a proof.



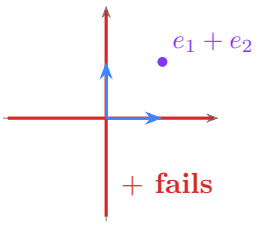
Line through 0



Shifted line



Quadrant



Two axes

EXAMPLE // Subspaces: the main ones

- V itself, and the trivial space $\{0\}$.
- A line through the origin in $\mathbb{R}^n$; a plane through the origin in $\mathbb{R}^3$.
- Convergent real sequences $\subseteq$ all real sequences (sum of convergent sequences converges, etc.).
- Continuous functions $I \to \mathbb{R} \subseteq$ all functions $I \to \mathbb{R}$.
- $\mathbb{R}_k[X] \subseteq \mathbb{R}_n[X]$ for $k < n$.

THEOREM / PROPOSITION // Subspaces are vector spaces

Any subspace U of a vector space V over $\mathbb{F}$ is itself a vector space over $\mathbb{F}$.

RECEIPTS (proof) //

The operations on U are those of V restricted to U. By definition $0_V \in U$, and U is stable under $+$ and $\cdot$, so the operations really land in U. All the algebraic identities (associativity, distributivity, ...) already hold in V, hence for elements of U too; e.g. $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$ because these are vectors of V. The zero of U is $0_V \in U$, and for $u \in U$, $-u = (-1)u \in U$ by stability under scalar multiplication. So U satisfies all the axioms. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Intersection of subspaces

If F and G are subspaces of E, then $F \cap G$ is a subspace of E.

RECEIPTS (proof) //

$0_E \in F$ and $0_E \in G$, so $0_E \in F \cap G$. Let $x, y \in F \cap G$, $\lambda \in \mathbb{F}$. Since $x, y \in F$ and F is a subspace, $x + y \in F$ and $\lambda x \in F$; same for G. Hence $x + y \in F \cap G$ and $\lambda x \in F \cap G$. ■ *(and that's on periodt)*

COMMON L (don't do this) // Union is NOT intersection

$F \cup G$ is generally **not** a subspace (see the "two axes" picture above: $(1,0) + (0,1) = (1,1)$ is on neither axis). Intersection: always fine. Union: only when one is inside the other.

THEOREM / PROPOSITION // A homogeneous linear equation defines a subspace

Let $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in \mathbb{R}$. Then $F = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n a_i x_i = 0\}$ is a subspace of $\mathbb{R}^n$.

RECEIPTS (proof) //

Zero: $\sum_{i=1}^{n} a_i \cdot 0 = 0$, so $0 \in F$. *Sum:* if $x, y \in F$, then $\sum_{i=1}^{n} a_i(x_i+y_i) = \sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{n} a_i y_i = 0 + 0 = 0$. *Scaling:* if $x \in F$, $\lambda \in \mathbb{R}$: $\sum_{i=1}^{n} a_i(\lambda x_i) = \lambda \sum_{i=1}^{n} a_i x_i = \lambda \cdot 0 = 0$. ■ *(and that's on periodt)*

3.2 Linear combinations and span

DEFINITION // Linear combination

Given $v_1, \dots, v_k \in \mathbb{R}^n$ and scalars $c_1, \dots, c_k \in \mathbb{R}$ (the **weights**), the vector

$$w = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = \sum_{i=1}^{k} c_i v_i$$

is a **linear combination** of $v_1, \dots, v_k$.

DEFINITION // Span (FR: sous-espace engendré, Vect)

$$\operatorname{span}\{v_1, \dots, v_k\} = \{c_1 v_1 + \cdots + c_k v_k \mid c_1, \dots, c_k \in \mathbb{R}\}.$$

It is *everything* you can cook with the ingredients $v_1, \dots, v_k$. French notation: $\operatorname{Vect}(v_1, \dots, v_k)$.

THEOREM / PROPOSITION // A span is always a subspace

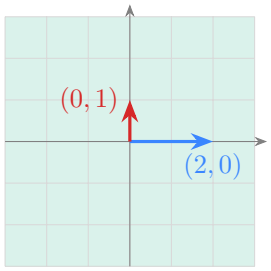
For any $v_1, \dots, v_k \in V$, $\operatorname{span}(v_1, \dots, v_k)$ is a subspace of V (and it's the *smallest* subspace containing all the v_i).

RECEIPTS (proof) //

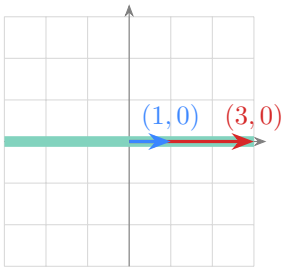
Zero: $0 = 0v_1 + \cdots + 0v_k$. *Sum:* $\sum_{i=1}^{k} c_i v_i + \sum_{i=1}^{k} c'_i v_i = \sum_{i=1}^{k} (c_i + c'_i) v_i$, again a linear combination. *Scaling:* $\lambda \sum_{i=1}^{k} c_i v_i = \sum_{i=1}^{k} (\lambda c_i) v_i$. Smallest: any subspace containing the v_i contains all their combinations by stability. ■ *(and that's on periodt)*

EXAMPLE // The three classic spans

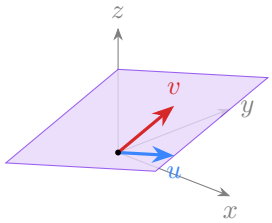
1. $\operatorname{span}((1,0),(0,1)) = \mathbb{R}^2$: any $(x, y) = x(1,0) + y(0,1)$.
2. $\operatorname{span}((2,0),(0,1)) = \mathbb{R}^2$ too: $(x, y) = \frac{x}{2}(2,0) + y(0,1)$. Stretching a vector doesn't shrink the span.
3. $\operatorname{span}((1,0),(3,0)) = \{(t,0) \mid t \in \mathbb{R}\}$, the x-axis only: $c_1(1,0) + c_2(3,0) = (c_1 + 3c_2, 0)$, second coordinate always 0. Two vectors, but only *one direction*. The second vector is redundant ($= 3\times$ the first).



span = $\mathbb{R}^2$ (whole plane)



span = the x-axis only



span$\{u, v\}$: a plane through 0 in $\mathbb{R}^3$

ML SIDE QUEST // Span = representation, dictionary learning

In **dictionary learning** we look for **atoms** $d_1, \dots, d_k \in \mathbb{R}^n$ (the dictionary $D = [d_1 \ \cdots \ d_k] \in \mathbb{R}^{n \times k}$) such that each data point is approximately a linear combination:

$$x \approx c_1 d_1 + \cdots + c_k d_k = Dc, \qquad c = (c_1, \dots, c_k)^\top.$$

So we want many data points to lie *close to* span$\{d_1, \dots, d_k\}$. Atoms = elementary patterns; coefficients = the recipe. Same idea powers dimensionality reduction and representation learning.

3.3 Lines, planes and affine subspaces

DEFINITION // Lines, planes, affine subspaces

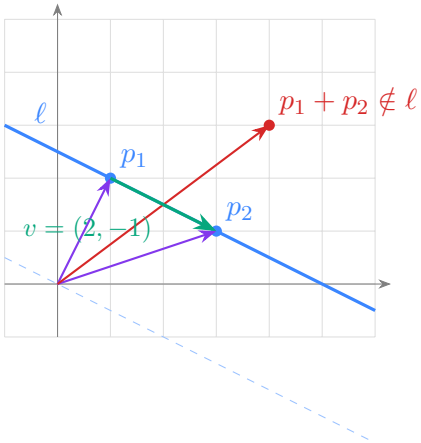
Line through p_0 with direction v: $\ell = \{p_0 + tv \mid t \in \mathbb{R}\}$. Plane through p_0 with directions u, v: $P = \{p_0 + su + tv \mid s, t \in \mathbb{R}\}$. In general, an **affine subspace** is a subspace translated by a fixed vector p_0:

$$P = \{p_0 + s_1 v_1 + \cdots + s_k v_k \mid s_1, \dots, s_k \in \mathbb{R}\}.$$

If $p_0 = 0$ you get back span$\{v_1, \dots, v_k\}$. If the v_i are linearly independent, P has dimension k.

EXAMPLE // Is $p_1 + p_2$ on the line?

$\ell = \{(1, 2) + t(2, -1) \mid t \in \mathbb{R}\}$. $p_1 = (1, 2)$ ($t = 0$) and $p_2 = (3, 1)$ ($t = 1$) are on ℓ. $p_1 + p_2 = (4, 3)$. Is there t with $(1 + 2t,\ 2 - t) = (4, 3)$? First coordinate: $t = 1.5$. Second: $2 - 1.5 = 0.5 \neq 3$. **No.** So ℓ is not closed under addition: an affine line not through 0 is **not** a subspace. (Also $0 \notin \ell$: $1 + 2t = 0 \Rightarrow t = -0.5$, then $2 - t = 2.5 \neq 0$.)



3.4 Spanning lists (familles génératrices)

DEFINITION // Spanning list

$(v_1, \dots, v_k)$ **spans** V (is a spanning list / *famille génératrice*) if $\text{span}(v_1, \dots, v_k) = V$, i.e. for every $x \in V$ there exist $\lambda_1, \dots, \lambda_k$ with $x = \sum_{i=1}^{k} \lambda_i v_i$.

EXAM CHEAT CODE // "Show that $(v_1, \dots, v_k)$ spans V"

1. Take an *arbitrary* $x \in V$ (with letters, e.g. $x = (a, b)$).
2. Write the unknown combination $\lambda_1 v_1 + \cdots + \lambda_k v_k = x$ as a linear system in the λ_i.
3. Solve it *in terms of* $a, b, \dots$. If a solution always exists: spanning. If for some x the system is impossible: not spanning.

EXAMPLE // Exercises, solved

(a) $\{(2,1),(0,1)\}$ **spans** $\mathbb{R}^2$**.** Let $(a,b) \in \mathbb{R}^2$. Solve $\lambda_1(2,1) + \lambda_2(0,1) = (a,b)$: $2\lambda_1 = a$ and $\lambda_1 + \lambda_2 = b$, so $\lambda_1 = \frac{a}{2}$, $\lambda_2 = b - \frac{a}{2}$. Always solvable, so it spans. Check with $(a,b) = (4,5)$: $\lambda_1 = 2, \lambda_2 = 3$ and $2(2,1) + 3(0,1) = (4,5)$ ✓.
(b) $V = \{(x,0,0) \mid x \in \mathbb{R}\}$ **and** $v_1 = (1,0,0)$**.** V is a subspace of $\mathbb{R}^3$: $(0,0,0) \in V$ ($x = 0$), $(x,0,0) + (y,0,0) = (x+y,0,0) \in V$, $\lambda(x,0,0) = (\lambda x, 0, 0) \in V$; so it's a vector space. And every $(x,0,0) = x \cdot v_1$, so (v_1) spans V.

3.5 Linear independence (familles libres)

DEFINITION // Linearly independent / free

$(v_1, \dots, v_k)$ is **linearly independent** if the only linear combination equal to 0 is the trivial one:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0 \implies \alpha_1 = \cdots = \alpha_k = 0.$$

Equivalently: *no vector of the family is a linear combination of the others.* Otherwise the family is **linearly dependent**.

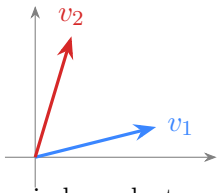
EXAM CHEAT CODE // "Is this family independent?"

1. Write $\alpha_1 v_1 + \cdots + \alpha_k v_k = 0$ with unknown α_i.
2. Turn it into a homogeneous linear system (one equation per coordinate).
3. Solve. If the *only* solution is $\alpha_1 = \cdots = \alpha_k = 0$: independent. If you find a non-zero solution: dependent, and **write that solution explicitly** as proof.

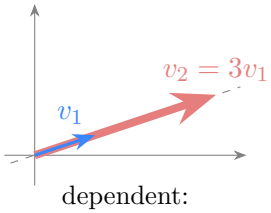
Speed checks: a family containing 0 is dependent; two vectors are dependent iff one is a multiple of the other; more than n vectors in $\mathbb{R}^n$ are always dependent.

EXAMPLE // Exercises, solved

(a) $v_1 = (1,0,0)$, $v_2 = (3,0,0)$, $v_3 = (0,0,1)$**.** Dependent: $3v_1 - v_2 + 0v_3 = (3,0,0) - (3,0,0) = 0$ with coefficients $(3,-1,0) \neq (0,0,0)$.
(b) $v_1 = (1,0)$, $v_2 = (0,2)$**.** $\alpha_1(1,0) + \alpha_2(0,2) = (\alpha_1, 2\alpha_2) = (0,0) \Rightarrow \alpha_1 = 0,\ \alpha_2 = 0$. Independent.



independent:
two genuinely different directions



dependent:
same line, v_2 adds nothing new

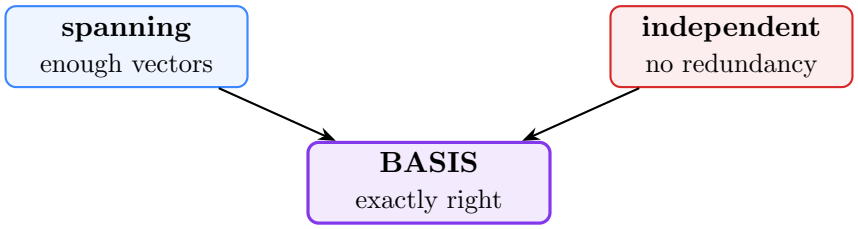
ML SIDE QUEST // Redundant features = linear dependence

Tabular data with x_1 = height in metres, x_2 = height in cm, x_3 = weight. Since $x_2 = 100x_1$, the feature columns are **linearly dependent**: x_2 brings no new direction. Linear regression hates this: the **design matrix** X (rows = data points, columns = features) becomes singular, i.e. there is $c \neq 0$ with $Xc = 0$ (here $c = (100, -1, 0)^\top$). Result: numerical instability, non-unique coefficients, poor generalisation. Fix: drop redundant features.

3.6 Basis and coordinates

DEFINITION // Basis

A **basis** of V is a family that is **linearly independent** AND **spanning**. The **canonical (standard) basis** of $\mathbb{R}^n$ is $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0)$, $\ldots$, $e_n = (0, \ldots, 0, 1)$.



too many vectors ⇒ can't be independent
too few vectors ⇒ can't span

THEOREM / PROPOSITION // Unique coordinates

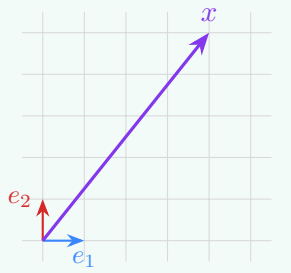
Given a basis $\mathcal{B} = (e_1, \ldots, e_n)$ of V, every $x \in V$ can be written in **exactly one** way as $x = \alpha_1 e_1 + \cdots + \alpha_n e_n$. The α_i are the **coordinates** of x in $\mathcal{B}$.

RECEIPTS (proof) // existence + uniqueness

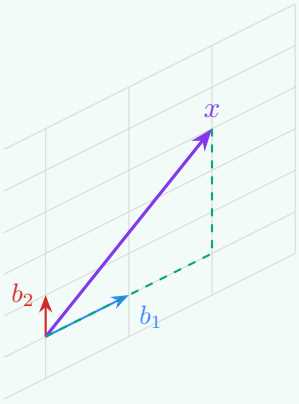
Existence. $\mathcal{B}$ spans V, so for every $x \in V$ there are scalars with $x = \sum_{i=1}^n \alpha_i e_i$.
Uniqueness. Suppose $x = \sum_{i=1}^n \alpha_i e_i = \sum_{i=1}^n \beta_i e_i$. Subtract: $0 = \sum_{i=1}^n (\alpha_i - \beta_i) e_i$. Since $\mathcal{B}$ is independent, $\alpha_i - \beta_i = 0$ for every $i = 1, \ldots, n$. So $\alpha_i = \beta_i$. ■ *(and that's on periodt)*

EXAMPLE // Same vector, different coordinates

$x = (4, 5)$. In the canonical basis: coordinates $(4, 5)$. In the basis $\mathcal{B}' = ((2, 1), (0, 1))$: we computed $x = 2(2, 1) + 3(0, 1)$, so coordinates $(2, 3)$. **Same arrow, different "address"**: coordinates depend on the basis.



canonical: $x = 4e_1 + 5e_2$



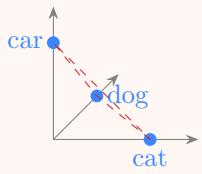
new basis: $x = 2b_1 + 3b_2$

ML SIDE QUEST // One-hot encoding = the canonical basis

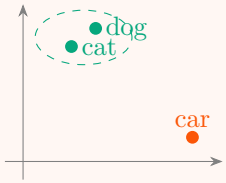
A categorical variable with K categories is encoded by $e_1, \dots, e_K \in \mathbb{R}^K$: category $j \mapsto e_j$. **One-hot encoding**. Every category gets its own axis; all categories are orthogonal and at the same distance $\sqrt{2}$ from each other: the encoding knows nothing about similarity.

ML SIDE QUEST // Embeddings: dense vectors whose geometry means something

One-hot vectors are sparse and huge (one axis per word of the vocabulary!). Instead, models learn **embeddings**: word $\mapsto x \in \mathbb{R}^d$ with d small. Image encoders map an image I to $z = \Phi(I) \in \mathbb{R}^d$. Goal: *similar things* $\Rightarrow$ *nearby vectors*. **Image retrieval** becomes a nearest-neighbour search: compute $z_q = \Phi(I_q)$, return images minimising $\|z_q - z_i\|_2$.



one-hot in $\mathbb{R}^3$: all equally far



embedding in $\mathbb{R}^2$: animals cluster

3.7 Dimension

THEOREM / PROPOSITION // Exchange lemma (Steinitz)

If $(v_1, \dots, v_p)$ is linearly independent in V and $(w_1, \dots, w_q)$ spans V, then $p \leq q$. *Independent families are never bigger than spanning families.*

RECEIPTS (proof) // swap them in one by one

Idea: starting from the spanning family $(w_1, \dots, w_q)$, replace w's by v's one at a time while staying spanning. There are only q slots, so we can't insert more than q vectors: $p \leq q$.
Step 1. Since the w_j span V, $v_1 = a_1 w_1 + \cdots + a_q w_q$. As $v_1 \neq 0$ (an independent family can't contain 0), some $a_j \neq 0$; reorder so $a_1 \neq 0$. Then $w_1 = \frac{1}{a_1} v_1 - \frac{a_2}{a_1} w_2 - \cdots - \frac{a_q}{a_1} w_q$, so everything generated by $(w_1, \dots, w_q)$ is generated by $(v_1, w_2, \dots, w_q)$: still spanning.
Step 2. Write v_2 as a combination of $v_1, w_2, \dots, w_q$. It can't use v_1 only (since (v_1, v_2) is independent), so some w_j has a non-zero coefficient: swap it for v_2, still spanning.

Continue. After inserting $v_1, \dots, v_r$ we still have a spanning family of q vectors. The process needs a w to swap at every step, so it can only go on while w's remain: $p \leq q$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Finite-dimensional vector spaces: the big 4

Let V be finite-dimensional (i.e. spanned by a finite family). Then:

1. V admits a basis.
2. All bases of V have the same number of elements.
3. Any linearly independent family can be **completed** into a basis (FR: *théorème de la base incomplète*).
4. From any spanning family you can **extract** a basis (FR: *théorème de la base extraite*).

RECEIPTS (proof) // the big 4

1. Existence. Take a finite spanning family $(v_1, \dots, v_m)$. If it's independent, it's a basis. If not, one vector is a combination of the others; remove it, the span doesn't change (any combination using it can be rewritten with the others). Repeat. The family is finite, so this stops, leaving a family that is spanning and independent: a basis.
2. Same size. Let $\mathcal{B} = (e_1, \dots, e_p)$, $\mathcal{C} = (f_1, \dots, f_q)$ be bases. $\mathcal{B}$ independent and $\mathcal{C}$ spanning $\Rightarrow p \leq q$ (exchange lemma). Swap roles $\Rightarrow q \leq p$. So $p = q$.
3. Completion. Let $(u_1, \dots, u_r)$ be independent. If it spans, done. Otherwise pick $v \notin \operatorname{span}(u_1, \dots, u_r)$. Claim $(u_1, \dots, u_r, v)$ is independent: if $a_1u_1 + \dots + a_ru_r + av = 0$ with $a \neq 0$, then $v = -\frac{a_1}{a}u_1 - \dots - \frac{a_r}{a}u_r \in \operatorname{span}(u_1, \dots, u_r)$, contradiction; so $a = 0$, and then all $a_i = 0$ by independence of the u_i. Keep adding. It must stop (an independent family has at most $\dim V$ vectors, by the exchange lemma), and it stops exactly when the family spans: a basis.
4. Extraction. Same as 1: remove redundant vectors from the spanning family until it's independent. ■ *(and that's on periodt)*

DEFINITION // Dimension

The common number of vectors in any basis of V is the **dimension** $\dim V$. $\dim \mathbb{R}^n = n$, $\dim\{0\} = 0$, $\dim \mathbb{R}_n[X] = n + 1$ (basis $1, X, \dots, X^n$), $\dim \mathcal{M}_{n,p}(\mathbb{R}) = np$.

EXAM CHEAT CODE // Dimension shortcuts (use these ALL the time)

In a space of dimension n:

- an independent family has **at most** n vectors; a spanning family has **at least** n vectors;
- **exactly** n **vectors**: independent $\iff$ spanning $\iff$ basis. So you only need to check *one* of the two properties!
- if $S \subseteq V$ is a subspace, $\dim S \leq \dim V$, with equality iff $S = V$.

EXAM CHEAT CODE // "Find a basis and the dimension of $S = \{x \mid \text{equations}\}$"

1. Solve the equations: express some variables (the *pivot* ones) in terms of the others (the *free* ones).
2. Write a general element of S with only the free variables, then split it: $x = \text{free}_1 \cdot v_1 + \text{free}_2 \cdot v_2 + \dots$

3. So $S = \text{span}(v_1, v_2, \dots)$. Check independence (usually immediate: look at the free-variable coordinates).
4. Basis $= (v_1, v_2, \dots)$, $\dim S =$ number of free variables $=$ (number of variables) $-$ (number of independent equations).

Example. $S = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$: $x_3 = -x_1 - x_2$, so $x = (x_1, x_2, -x_1 - x_2) = x_1(1, 0, -1) + x_2(0, 1, -1)$. Basis $((1, 0, -1), (0, 1, -1))$ (independent: look at the first two coordinates), $\dim S = 2 = 3 - 1$.

DEFINITION // Rank of a family

$\text{rank}(v_1, \dots, v_k) = \dim \text{span}(v_1, \dots, v_k)$: the number of genuinely different directions in the family.

ML SIDE QUEST // Rank of data = how many real directions

Redundant features: for $X = [x_1\ x_2\ x_3] \in \mathbb{R}^{m\times 3}$ with $x_2 = 100x_1$, the columns x_1, x_2 span only one direction, so $\text{rank}(X) \leq 2 < 3$. **Rank reveals linear redundancy.**
Face images: a dataset $X \in \mathbb{R}^{m\times p}$ (one face per row, p pixels) has p huge, but faces vary mainly along a few factors (lighting, pose, expression, identity). So $\text{rank}(X) \ll \min(m, p)$, at least approximately: *low-dimensional structure inside a high-dimensional pixel space.*

3.8 Sums and direct sums of subspaces

DEFINITION // Sum, direct sum

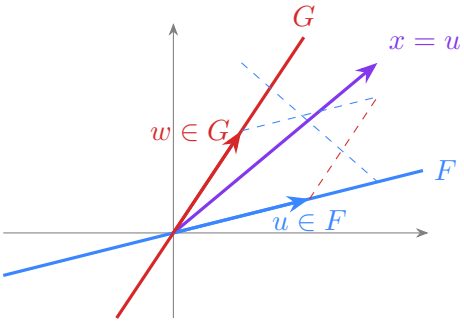
For subspaces F, G of E: $F + G = \{x \in E \mid \exists u \in F,\ \exists w \in G,\ x = u + w\}$. If $F \cap G = \{0\}$, the sum is **direct**, written $F \oplus G$. If moreover $F \oplus G = E$, F and G are **supplementary** (FR: *supplémentaires*).

THEOREM / PROPOSITION // Sum of subspaces + uniqueness in a direct sum

1. $F + G$ is a subspace of E (the smallest one containing F and G).
2. If $F \cap G = \{0\}$, every $x \in F + G$ can be written **uniquely** as $x = u + w$ with $u \in F$, $w \in G$.

RECEIPTS (proof) //

1. $0_E = 0_E + 0_E$ with $0_E \in F$, $0_E \in G$. If $x = u + w$ and $y = u' + w'$ $(u, u' \in F,\ w, w' \in G)$, then $x + y = (u + u') + (w + w')$ with $u + u' \in F$, $w + w' \in G$. And $\lambda x = \lambda u + \lambda w$ with $\lambda u \in F$, $\lambda w \in G$. So $F + G$ is a subspace.
2. Existence is the definition of $F + G$. Uniqueness: if $x = u + w = u' + w'$, then $u - u' = w' - w$. The left side is in F, the right side in G, so this common vector is in $F \cap G = \{0\}$. Hence $u = u'$ and $w = w'$. ■ *(and that's on periodt)*



$\mathbb{R}^2 = F \oplus G$: two different lines. Every x splits in exactly one way into a piece on F + a piece on G.

Example: $F = \text{span}(e_1)$, $G = \text{span}(e_2)$, $(4, 5) = (4, 0) + (0, 5)$.

THEOREM / PROPOSITION // Grassmann dimension formula

For finite-dimensional subspaces U, W of V:

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Corollary: for a direct sum, $\dim(U \oplus W) = \dim U + \dim W$.

Inclusion–exclusion energy: you count the intersection twice, so subtract it once. Proof in Section 6 (it uses rank–nullity).

EXAMPLE // Two planes in $\mathbb{R}^3$

Two distinct planes through 0 in $\mathbb{R}^3$: $\dim U = \dim W = 2$, and $U + W = \mathbb{R}^3$ (it's bigger than a plane). So $\dim(U \cap W) = 2 + 2 - 3 = 1$: **two distinct planes through the origin always meet in a line.** E.g. $U = \{z = 0\}$, $W = \{x = 0\}$ meet along the y-axis.

4 Functions: injective, surjective, bijective (the deep dive)

TL;DR // The three words everyone mixes up

Injective = no two inputs share an output ("no collisions"). **Surjective** = every output is hit ("nobody left out"). **Bijective** = both = a perfect one-to-one pairing ("everyone has exactly one match"), so it can be undone.

4.1 Vocabulary

DEFINITION // Function

A **function** (or **map**) f from a set X to a set Y assigns to **each** element of X **exactly one** element of Y. Notation $f : X \to Y$, $x \mapsto f(x)$.

- X = **domain**, Y = **codomain**.
- $y = f(x)$: f *maps* x to y; y is the **image** of x; x is the **argument**/variable.

DEFINITION // Image (range) and preimage

Image/range: $f(X) = \{f(x) \mid x \in X\} \subseteq Y$, all outputs actually produced. For $A \subseteq X$, $f(A) = \{f(a) \mid a \in A\} \subseteq f(X)$.
Preimage of $y \in Y$: $f^{-1}(y) = \{x \in X \mid f(x) = y\}$, all inputs that land on y. (It is a *set*, possibly empty, possibly huge. The notation $f^{-1}(y)$ makes sense even if f has no inverse!)

EXAMPLE // Image and preimage for $f(x) = x^2$, $f : \mathbb{R} \to \mathbb{R}$

Image: $f(\mathbb{R}) = [0, +\infty)$. Preimages: $f^{-1}(4) = \{-2, 2\}$ (two elements), $f^{-1}(0) = \{0\}$ (one element), $f^{-1}(-1) = \emptyset$ (none).

4.2 The three definitions

Let $f : X \to Y$.

DEFINITION // Injective (one-to-one)

f is **injective** if different inputs always give different outputs:

$$\forall a, b \in X, \quad a \neq b \;\Rightarrow\; f(a) \neq f(b) \qquad \Longleftrightarrow \qquad \forall a, b \in X, \quad f(a) = f(b) \;\Rightarrow\; a = b.$$

Equivalently: every $y \in Y$ has **at most one** preimage, $|f^{-1}(y)| \leq 1$.

DEFINITION // Surjective (onto)

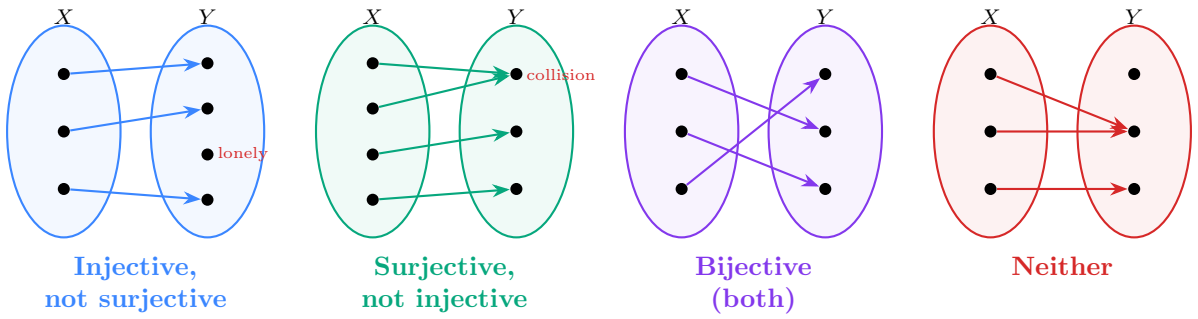
f is **surjective** if its image is the whole codomain, $f(X) = Y$:

$$\forall y \in Y, \; \exists x \in X, \quad f(x) = y.$$

Equivalently: every $y \in Y$ has **at least one** preimage, $f^{-1}(y) \neq \emptyset$.

DEFINITION // Bijective

f is **bijective** if it is injective **and** surjective: every $y \in Y$ has **exactly one** preimage. Then f has an inverse $f^{-1} : Y \to X$ with $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.



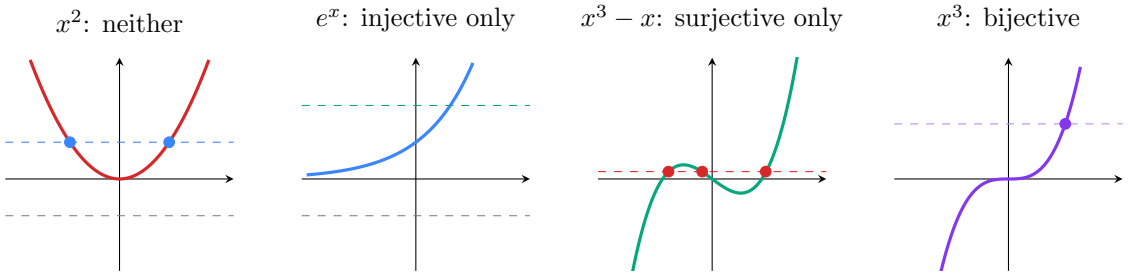
COMMON L (don't do this) // Things that are NOT the definition

- "Every x has one image" is just the definition of a *function*, not injectivity.
- Injective is about *outputs not being shared*. Surjective is about *outputs all being used*. They are independent properties: all four combos exist (see picture).
- Surjectivity depends on the **codomain**! $x \mapsto x^2$ is not surjective onto $\mathbb{R}$, but it is surjective onto $[0, +\infty)$. Always say "surjective onto ...".
- Injectivity depends on the **domain**: $x \mapsto x^2$ is not injective on $\mathbb{R}$, but it is on $[0, +\infty)$.

4.3 The graph test (for $f : \mathbb{R} \to \mathbb{R}$)

Draw horizontal lines $y = c$ for every c in the codomain:

- **injective** $\iff$ every horizontal line hits the graph **at most once**;
- **surjective** $\iff$ every horizontal line hits the graph **at least once**;
- **bijective** $\iff$ every horizontal line hits the graph **exactly once**.



All as maps $\mathbb{R} \to \mathbb{R}$. x^2: the line $y = 1$ hits twice (not injective), $y = -1$ never (not surjective). e^x: never twice, but $y = -1$ is never hit. $x^3 - x$: $y = 0.2$ hit three times but every line is hit at least once. x^3: exactly once, always.

4.4 How to prove it: the recipes

EXAM CHEAT CODE // "Prove f is injective"

1. Write: "Let $a, b \in X$ such that $f(a) = f(b)$."
2. Do algebra on $f(a) = f(b)$ until you get $a = b$.
3. Conclude "hence f is injective."

For a linear map: just show $\ker f = \{0\}$, i.e. "let x with $f(x) = 0$, show $x = 0$". (Proof of why that's enough in Section 5.)

EXAM CHEAT CODE // "Prove f is NOT injective"

Find two **explicit** inputs $a \neq b$ with $f(a) = f(b)$. For a linear map: find one explicit $x \neq 0$ with $f(x) = 0$. Or use dimensions: if dim(domain) $>$ dim(codomain), a linear map can't be injective.

EXAM CHEAT CODE // "Prove f is surjective"

1. Write: "Let $y \in Y$." (arbitrary!)
2. **Solve** $f(x) = y$ for x (scratch work), find a formula $x = \ldots$ in terms of y.
3. Check that this x is in X and that $f(x) = y$ really holds.
4. Conclude.

For a linear map between finite-dim spaces: show $\operatorname{rank} f = \dim Y$.

EXAM CHEAT CODE // "Prove f is NOT surjective"

Find one explicit $y \in Y$ that is never reached (show $f(x) = y$ has no solution). For a linear map: if dim(domain) $<$ dim(codomain), it can't be surjective.

EXAM CHEAT CODE // "Prove f is bijective"

Either (i) prove injective + surjective, or (ii) exhibit an inverse $g : Y \to X$ and check $g(f(x)) = x$ for all x and $f(g(y)) = y$ for all y. For linear maps with $\dim X = \dim Y$: injective alone is enough (or surjective alone)!

EXAMPLE // Full worked examples

1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 2$ **is bijective.** *Injective:* let $a, b \in \mathbb{R}$ with $3a + 2 = 3b + 2$. Then $3a = 3b$, so $a = b$. *Surjective:* let $y \in \mathbb{R}$. Solve $3x + 2 = y$: $x = \frac{y-2}{3} \in \mathbb{R}$, and $f\left(\frac{y-2}{3}\right) = y - 2 + 2 = y$ ✓. Inverse: $f^{-1}(y) = \frac{y-2}{3}$.
2. $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^2$ **is neither.** Not injective: $g(-2) = g(2) = 4$ but $-2 \neq 2$. Not surjective: $y = -1$ has no preimage since $x^2 \geq 0$.
3. $h : [0, +\infty) \to [0, +\infty)$, $h(x) = x^2$ **is bijective.** Same formula, new domain/codomain! If $a^2 = b^2$ with $a, b \geq 0$ then $a = b$; for $y \geq 0$, $x = \sqrt{y}$ works. *Moral: the domain and codomain are part of the function's identity.*
4. $\exp : \mathbb{R} \to \mathbb{R}$ **is injective, not surjective**; $\exp : \mathbb{R} \to (0, +\infty)$ is bijective, with inverse ln.
5. $p : \mathbb{R}^2 \to \mathbb{R}$, $p(x, y) = x$ **is surjective, not injective.** Surjective: for $t \in \mathbb{R}$, $p(t, 0) = t$. Not injective: $p(1, 0) = p(1, 5) = 1$.
6. $\iota : \mathbb{R} \to \mathbb{R}^2$, $\iota(t) = (t, 0)$ **is injective, not surjective.** $\iota(a) = \iota(b) \Rightarrow (a, 0) = (b, 0) \Rightarrow a = b$. $(0, 1)$ is never reached.

4.5 Useful facts

THEOREM / PROPOSITION // Composition and finite sets

Let $f : X \to Y$, $g : Y \to Z$.

1. If f and g are injective, so is $g \circ f$. If both are surjective, so is $g \circ f$.
2. If $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.
3. (Pigeonhole) If X, Y are finite: f injective $\Rightarrow |X| \leq |Y|$; f surjective $\Rightarrow |X| \geq |Y|$; bijective $\Rightarrow |X| = |Y|$.

RECEIPTS (proof) // a couple of them

1 (injective). If $g(f(a)) = g(f(b))$, injectivity of g gives $f(a) = f(b)$, then injectivity of f gives $a = b$. *1 (surjective).* Let $z \in Z$. As g is surjective, $z = g(y)$ for some y; as f is surjective, $y = f(x)$ for some x. So $z = g(f(x))$. *2.* If $f(a) = f(b)$ then $g(f(a)) = g(f(b))$, so $a = b$ by injectivity of $g \circ f$. ■ *(and that's on periodt)*

The linear-algebra version of the pigeonhole principle is rank–nullity: dimensions play the role of the number of elements. Coming up.

5 Linear maps: functions that respect the vibe

TL;DR // Chapter 5 in one breath

A linear map is a function between vector spaces that respects + and scaling. It sends lines to lines, keeps the origin fixed, and is fully determined by what it does to a basis. Its **kernel** is what gets crushed to 0 (information lost); its **range** is what it can output.

5.1 Definition and vocabulary

DEFINITION // Linear map

$L : U \to V$ (U, V vector spaces) is **linear** if for all $u_1, u_2 \in U$ and all $a, b \in \mathbb{R}$:

$$L(au_1 + bu_2) = aL(u_1) + bL(u_2).$$

Equivalently: $L(u_1 + u_2) = L(u_1) + L(u_2)$ **and** $L(au) = aL(u)$.

DEFINITION // The fancy names

- **Homomorphism**: structure-preserving map between two structures of the same type. For vector spaces: *linear map / linear transformation.*
- **Endomorphism**: linear map from V to itself.
- **Isomorphism**: bijective linear map. U and V are **isomorphic** if there's an isomorphism $U \to V$ (then its inverse $V \to U$ is one too).

THEOREM / PROPOSITION // First consequences

If L is linear then $L(0_U) = 0_V$ and $L\left(\sum_{i=1}^{k} c_i u_i\right) = \sum_{i=1}^{k} c_i L(u_i)$.

RECEIPTS (proof) //

$L(0_U) = L(0_U + 0_U) = L(0_U) + L(0_U)$; subtract $L(0_U)$ from both sides: $L(0_U) = 0_V$. The second claim follows by induction on k from the definition. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Show L is linear"

1. Let $u, u' \in U$, $a, b \in \mathbb{R}$ (arbitrary).
2. Compute the LEFT side $L(au + bu')$ by plugging $au + bu'$ into the formula.
3. Compute the RIGHT side $aL(u) + bL(u')$ separately.
4. Show they're equal (usually by expanding/regrouping).

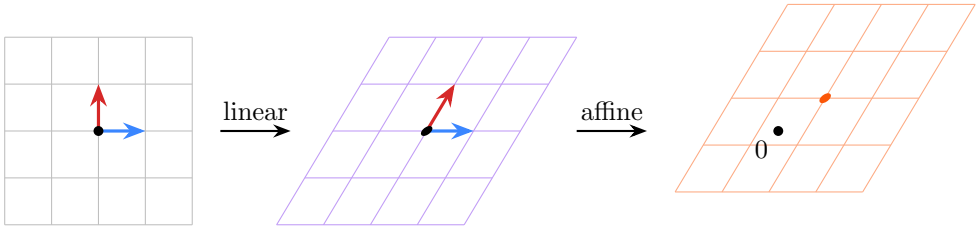
Shortcuts: each output coordinate is of the form $c_1x_1 + \cdots + c_nx_n$ (no constants, no squares, no products of variables) $\Rightarrow$ linear. Matrix times vector $x \mapsto Wx$ is linear. Compositions and sums of linear maps are linear.

EXAM CHEAT CODE // "Show L is NOT linear"

Fastest: check $L(0) \neq 0$ (kills any map with a constant term, like $x \mapsto Wx + b$ with $b \neq 0$). Otherwise find explicit u and a with $L(au) \neq aL(u)$ (e.g. $L(x) = x^2$: $L(2 \cdot 1) = 4 \neq 2 = 2L(1)$).

EXAMPLE // Exercises, solved

(a) $L(x,y) = (x,0)$ **is linear.** Let $u = (x,y)$, $u' = (x',y')$, $a,b \in \mathbb{R}$. Then $au + bu' = (ax + bx', ay+by')$, so $L(au+bu') = (ax+bx', 0)$. And $aL(u)+bL(u') = a(x,0)+b(x',0) = (ax+bx',0)$. Equal ✓.
(b) $L(x,y) = 10\cdot(x,y)$ **is linear.** $L(au+bu') = 10(au+bu') = a(10u)+b(10u') = aL(u)+bL(u')$ by distributivity ✓.
(c) Non-examples. $T(x,y) = (x+1,y)$: $T(0,0) = (1,0) \neq 0$, not linear. $Q(x,y) = xy$: $Q(2(1,1)) = 4 \neq 2 = 2Q(1,1)$, not linear.



input grid — lines stay lines, parallel stays parallel, origin stays put — $x \mapsto Ax + b$: origin moved ⇒ NOT linear

ML SIDE QUEST // Neural network layers

A dense layer *without bias and without activation* is $L : \mathbb{R}^d \to \mathbb{R}^k$, $L(x) = Wx$ with $W \in \mathbb{R}^{k\times d}$: **linear**. With bias, $x \mapsto Wx + b$ is **affine**, not linear unless $b = 0$ (it moves the origin). The range $\text{range}(L) = \{Wx \mid x \in \mathbb{R}^d\}$ is the set of all outputs the layer can ever produce.

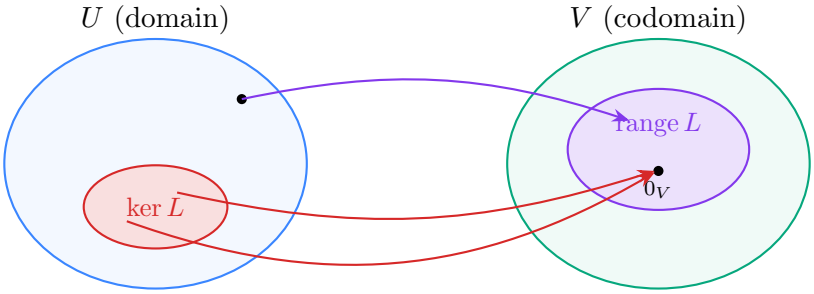
5.2 Kernel and range

DEFINITION // Kernel (null space) and range (image)

For $L : U \to V$ linear:

$$\ker(L) = \{x \in U \mid L(x) = 0_V\} \subseteq U, \qquad \text{range}(L) = \{y \in V \mid \exists x \in U,\ L(x) = y\} = L(U) \subseteq V.$$

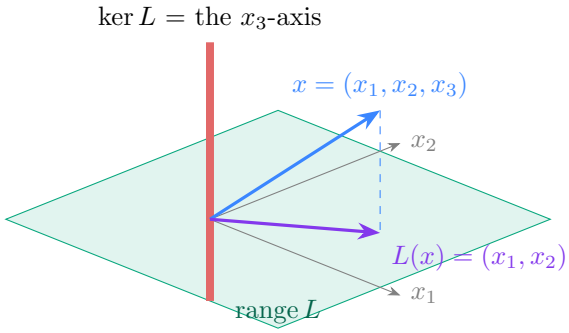
Also written $\text{null}(L)$ and $\text{Im}(L)$. *Warning: "kernel" means something else in ML (kernel methods, SVM kernels, conv kernels). Context is everything.*



the whole kernel collapses onto 0_V;
the range is everything that gets hit

EXAMPLE // Kernel and range, by hand

1. $L(x,y) = x$, $L : \mathbb{R}^2 \to \mathbb{R}$. $\ker L = \{(x,y) \mid x = 0\} = \{(0,y) \mid y \in \mathbb{R}\}$ (the y-axis). $\operatorname{range} L = \mathbb{R}$ (for $t \in \mathbb{R}$, $L(t,0) = t$).
2. Projection $L : \mathbb{R}^3 \to \mathbb{R}^2$, $L(x_1,x_2,x_3) = (x_1,x_2)$. $\ker L = \{(0,0,t) \mid t \in \mathbb{R}\}$ (the vertical axis), $\operatorname{range} L = \mathbb{R}^2$.
3. $L(x,y) = (2x,0)$, $L : \mathbb{R}^2 \to \mathbb{R}^2$. $\ker L = \{(0,y)\}$, $\operatorname{range} L = \{(s,0) \mid s \in \mathbb{R}\}$ (the x-axis).



Projection $\mathbb{R}^3 \to \mathbb{R}^2$:
the height x_3 is **lost**.
Every vertical line collapses
to a single point.

THEOREM / PROPOSITION // Kernel, range, injectivity, surjectivity

If $L : U \to V$ is linear, then

1. $\ker(L)$ is a subspace of U;
2. $\operatorname{range}(L)$ is a subspace of V;
3. L is injective $\iff$ $\ker(L) = \{0_U\}$;
4. L is surjective $\iff$ $\operatorname{range}(L) = V$.

RECEIPTS (proof) //

1. $L(0_U) = 0_V$ so $0_U \in \ker L$. If $x, y \in \ker L$: $L(x+y) = L(x) + L(y) = 0_V + 0_V = 0_V$. If $x \in \ker L$, $\lambda \in \mathbb{F}$: $L(\lambda x) = \lambda L(x) = \lambda 0_V = 0_V$.
2. $0_V = L(0_U) \in \operatorname{range} L$. If $y_1 = L(x_1)$, $y_2 = L(x_2)$: $y_1 + y_2 = L(x_1 + x_2) \in \operatorname{range} L$ and $\lambda y_1 = L(\lambda x_1) \in \operatorname{range} L$.
3. *($\Rightarrow$)* Suppose L injective, and let $x \in \ker L$. Then $L(x) = 0_V = L(0_U)$, so $x = 0_U$ by injectivity. Hence $\ker L = \{0_U\}$. *($\Leftarrow$)* Suppose $\ker L = \{0_U\}$ and $L(x) = L(y)$. Then $L(x-y) = L(x) - L(y) = 0_V$, so $x - y \in \ker L$, i.e. $x - y = 0_U$, so $x = y$. Hence injective.
4. Surjective means every $v \in V$ is some $L(u)$: that's literally $\operatorname{range} L = V$. ■ *(and that's on periodt)*

TL;DR // Why point 3 is a cheat code

For a general function, injectivity means checking *all pairs* (a,b). For a *linear* map, you only need to check what goes to 0. Linearity turns "$f(a) = f(b)$" into "$f(a-b) = 0$". One collision anywhere $\iff$ a non-zero vector in the kernel.

THEOREM / PROPOSITION // Range of the zero map

$\operatorname{range}(L) = \{0_V\}$ $\iff$ $L = 0$ (the map sending everything to 0_V).

RECEIPTS (proof) //

($\Leftarrow$) If $L = 0$, the only output is 0_V. ($\Rightarrow$) If range $L = \{0_V\}$, then for every $x \in U$, $L(x) \in \text{range } L$ so $L(x) = 0_V$: L is the zero map. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Determine $\ker L$" (with a basis and its dimension)

1. Write $L(x) = 0$ as a system of equations (one per output coordinate).
2. Solve it: pivot variables in terms of free variables.
3. Write a general solution as $\text{free}_1 \cdot v_1 + \text{free}_2 \cdot v_2 + \dots$
4. $\ker L = \text{span}(v_1, v_2, \dots)$; the v_i are a basis; $\dim \ker L =$ number of free variables.
5. Injective iff you found no free variable (only $x = 0$).

EXAM CHEAT CODE // "Determine range L"

1. $\text{range } L = \text{span}(L(e_1), \dots, L(e_n))$ for any basis (e_i) of U (images of a basis span the range).
2. Extract a basis from these vectors (remove the redundant ones).
3. Or: compute $\dim \text{range } L = \dim U - \dim \ker L$ (rank–nullity) and use it.

5.3 A linear map is determined by a basis

THEOREM / PROPOSITION // Define it on a basis, get it everywhere

Let $\mathcal{B}_U = (u_1, \dots, u_n)$ be a basis of U and $\mathcal{F}_V = (v_1, \dots, v_n)$ *any* family in V. There exists a **unique** linear map $g : U \to V$ with $g(u_i) = v_i$ for all $1 \le i \le n$. Moreover:

1. g injective $\iff$ $\mathcal{F}_V$ linearly independent;
2. g surjective $\iff$ $\mathcal{F}_V$ spans V;
3. g bijective $\iff$ $\mathcal{F}_V$ is a basis of V.

RECEIPTS (proof) // construct, uniqueness, then the 3 properties

Existence. Any $x \in U$ has unique coordinates $x = \sum_{i=1}^n \alpha_i u_i$; define $g(x) = \sum_{i=1}^n \alpha_i v_i$ (well defined since coordinates are unique). Linearity: if $y = \sum_{i=1}^n \beta_i u_i$ then $\lambda x + \mu y = \sum_{i=1}^n (\lambda\alpha_i + \mu\beta_i)u_i$, so $g(\lambda x + \mu y) = \sum_{i=1}^n (\lambda\alpha_i + \mu\beta_i)v_i = \lambda g(x) + \mu g(y)$. And u_j has coordinates 1 in position j, 0 elsewhere, so $g(u_j) = v_j$.
Uniqueness. If h is linear with $h(u_i) = v_i$: $h(x) = \sum_{i=1}^n \alpha_i h(u_i) = \sum_{i=1}^n \alpha_i v_i = g(x)$.
Injectivity. If (v_i) is independent and $g(x) = 0$ with $x = \sum \alpha_i u_i$, then $\sum \alpha_i v_i = 0$, so all $\alpha_i = 0$, $x = 0$: $\ker g = \{0\}$. Conversely if (v_i) is dependent, take α's not all zero with $\sum \alpha_i v_i = 0$; then $x = \sum \alpha_i u_i \neq 0$ (the u_i are independent) but $g(x) = 0$: not injective.
Surjectivity. $\text{range } g = \{\sum \alpha_i v_i\} = \text{span}(v_1, \dots, v_n)$, so g surjective iff $\text{span}(v_i) = V$.
Bijectivity. Both at once: independent and spanning, i.e. a basis. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Isomorphic spaces

Two finite-dimensional vector spaces over the same field with the **same dimension** are isomorphic.

RECEIPTS (proof) //

Let $\dim U = \dim V = n$, (u_i) a basis of U, (v_i) a basis of V. By the previous theorem there's a unique linear g with $g(u_i) = v_i$, and since (v_i) is a basis, g is bijective: an isomorphism. ■ *(and that's on periodt)*

So every n-dimensional real vector space is "$\mathbb{R}^n$ in disguise": polynomials of degree ≤ 2 are basically $\mathbb{R}^3$ via $a_0 + a_1X + a_2X^2 \mapsto (a_0, a_1, a_2)$. Coordinates are the isomorphism.

ML SIDE QUEST // Kernel = lost information

For $L : \mathbb{R}^d \to \mathbb{R}^k$, vectors in $\ker L$ are directions **completely collapsed** by L: if $x' = x + n$ with $n \in \ker L$, then $L(x') = L(x)$, so the model *cannot tell x and x' apart*. In dimensionality reduction, the kernel is the discarded information. In PCA, the kernel of the projection is spanned by the directions of *least variance*, which we choose to throw away.

6 Rank and the rank–nullity theorem

TL;DR // Chapter 6 in one breath

Rank = dimension of what comes out. Nullity = dimension of what gets crushed. **Rank + nullity = dimension of what goes in.** Every input direction is either kept or destroyed, nothing disappears from the accounting.

DEFINITION // Rank of a linear map

$\operatorname{rank}(L) = \dim(\operatorname{range} L) = \dim L(U)$, where $L(U) = \{y \in V \mid y = L(x),\ x \in U\}$. Also $\operatorname{rank} L = \dim \operatorname{Im} L$.

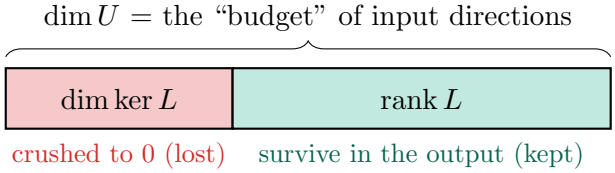
EXAMPLE // Ranks

$L : \mathbb{R}^2 \to \mathbb{R}$, $L(x, y) = x$ has rank 1 (range $= \mathbb{R}$). The zero map has rank 0. The identity of $\mathbb{R}^n$ has rank n. Always: $\operatorname{rank} L \leq \dim V$ (the range lives in V) and $\operatorname{rank} L \leq \dim U$ (the range is spanned by the images of a basis of U).

THEOREM / PROPOSITION // Rank–nullity theorem

If U, V are finite-dimensional and $L : U \to V$ is linear, then

$$\boxed{\operatorname{rank}(L) + \dim(\ker L) = \dim(U)}$$



RECEIPTS (proof) // complete a basis of the kernel

Let $p = \dim \ker L$ and $(e_1, \ldots, e_p)$ a basis of $\ker L$. It's independent in U, so complete it into a basis $(e_1, \ldots, e_p, u_1, \ldots, u_q)$ of U; thus $\dim U = p + q$. Claim: $(L(u_1), \ldots, L(u_q))$ is a basis of range L.

Spanning. Let $y = L(x) \in \operatorname{range} L$ with $x = \sum_{i=1}^{p} \alpha_i e_i + \sum_{j=1}^{q} \beta_j u_j$. Then $y = \sum_{i=1}^{p} \alpha_i L(e_i) + \sum_{j=1}^{q} \beta_j L(u_j) = \sum_{j=1}^{q} \beta_j L(u_j)$ because $L(e_i) = 0$.
Independence. If $\sum_{j=1}^{q} \beta_j L(u_j) = 0$, then $L\big(\sum_j \beta_j u_j\big) = 0$, so $\sum_j \beta_j u_j \in \ker L$ and $\sum_{j=1}^{q} \beta_j u_j = \sum_{i=1}^{p} \alpha_i e_i$ for some α_i. Rearranging, $\sum_i (-\alpha_i) e_i + \sum_j \beta_j u_j = 0$; this is a basis of U, so all coefficients vanish, in particular $\beta_1 = \cdots = \beta_q = 0$.
So $\operatorname{rank} L = q$, and $\operatorname{rank} L + \dim \ker L = q + p = \dim U$. ■ *(and that's on periodt)*

EXAMPLE // Rank–nullity in action

1. $L : \mathbb{R}^2 \to \mathbb{R}^2$, $(x, y) \mapsto (2x, 0)$. $\ker L = \{(0, y)\}$, dim 1; $\operatorname{range} L = \{(s, 0)\}$, dim 1. Check: $1 + 1 = 2 = \dim \mathbb{R}^2$ ✓.
2. $L : \mathbb{R}^3 \to \mathbb{R}^2$, $(x_1, x_2, x_3) \mapsto (x_1, x_2)$. $\operatorname{rank} L = 2$, $\ker L = \{(0, 0, t)\}$, dim 1. Check: $2 + 1 = 3$ ✓. *The third coordinate is discarded.*
3. $L : \mathbb{R}^3 \to \mathbb{R}$, $L(x) = x_1 + x_2 + x_3$. Range $= \mathbb{R}$ ($L(t, 0, 0) = t$), so rank 1. Then without solving anything: $\dim \ker L = 3 - 1 = 2$. (Matches the plane $x_1 + x_2 + x_3 = 0$ found earlier.)

THEOREM / PROPOSITION // Rank vs injective / surjective / bijective

For $L : U \to V$ linear, U, V finite-dimensional:

1. L injective $\iff \operatorname{rank} L = \dim U$;
2. L surjective $\iff \operatorname{rank} L = \dim V$;
3. L bijective $\iff \operatorname{rank} L = \dim U = \dim V$.

RECEIPTS (proof) //

1. Injective $\iff \ker L = \{0\} \iff \dim \ker L = 0 \iff \operatorname{rank} L = \dim U$ (rank–nullity). *2.* Surjective $\iff \operatorname{range} L = V \iff \dim \operatorname{range} L = \dim V$ (a subspace with the full dimension is the whole space). *3.* Combine 1 and 2. ■ *(and that's on periodt)*

EXAM CHEAT CODE // The dimension-counting shortcuts (exam gold)

For $L : U \to V$ linear, $n = \dim U$, $m = \dim V$:

Situation	Conclusion	Why
$n > m$ (squashing)	**never injective**	$\dim \ker L = n - \operatorname{rank} L \geq n - m > 0$
$n < m$ (embedding)	**never surjective**	$\operatorname{rank} L \leq n < m$
$n = m$	inj. $\iff$ surj. $\iff$ bij.	rank–nullity

RECEIPTS (proof) // Grassmann formula $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$

Define $\varphi : U \times W \to U + W$, $\varphi(u, w) = u + w$.
Linear: $\varphi(\lambda(u, w) + \mu(u', w')) = (\lambda u + \mu u') + (\lambda w + \mu w') = \lambda\varphi(u, w) + \mu\varphi(u', w')$.
Range: every element of $U + W$ is some $u + w$, so $\operatorname{range} \varphi = U + W$ and $\operatorname{rank} \varphi = \dim(U + W)$.
Kernel: $\varphi(u, w) = 0 \iff w = -u$. Then $u \in U$ and $u = -w \in W$, so $u \in U \cap W$. Hence $\ker \varphi = \{(z, -z) \mid z \in U \cap W\}$, and $z \mapsto (z, -z)$ is an isomorphism $U \cap W \to \ker \varphi$, so $\dim \ker \varphi = \dim(U \cap W)$.
Rank–nullity: $\dim(U + W) + \dim(U \cap W) = \dim(U \times W) = \dim U + \dim W$. Rearrange.
Corollary: for a direct sum $U \cap W = \{0\}$, so $\dim(U \oplus W) = \dim U + \dim W$. ■ *(and that's on periodt)*

ML SIDE QUEST // Rank of a data matrix

$X \in \mathbb{R}^{m\times d}$: m observations, d features. $\operatorname{rank} X$ = number of independent linear directions present in the data.

- If $\operatorname{rank} X < d$ (with $d < m$): some features are linear combinations of others (redundancy).
- If $m < d$ (more features than data points!): $\operatorname{rank} X \le m$, the data lie in a subspace of $\mathbb{R}^d$ of dimension at most m.

ML SIDE QUEST // Rank of a linear layer, and LoRA

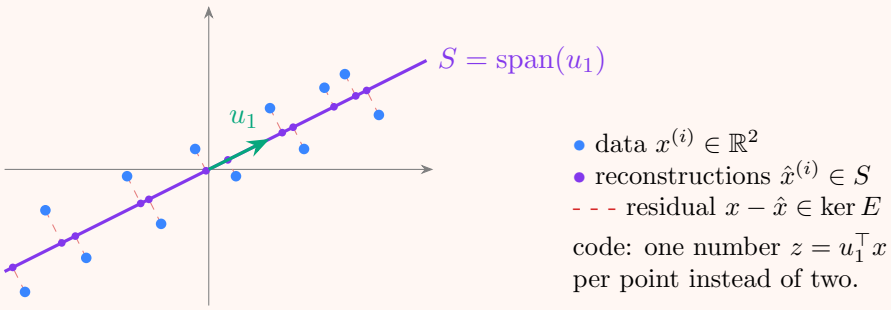
For $L(x) = Wx$, $W \in \mathbb{R}^{k\times d}$: $\operatorname{rank} W = \dim \operatorname{range} L$ = dimension of outputs actually reachable. If $\operatorname{rank} W < k$ the layer *cannot* produce all vectors of $\mathbb{R}^k$.
LoRA (Low-Rank Adaptation) fine-tunes a big model by adding $\Delta W = BA$ with $B \in \mathbb{R}^{k\times r}$, $A \in \mathbb{R}^{r\times d}$, r tiny. Then $\operatorname{rank}(\Delta W) \le r$: you train $r(k+d)$ numbers instead of kd. The rank r controls how much the adaptation can express.

ML SIDE QUEST // PCA as compression: the whole theory in one algorithm

Let $x^{(1)}, \dots, x^{(m)} \in \mathbb{R}^d$ be centred data ($\bar{x} = 0$). PCA finds orthonormal directions $u_1, \dots, u_k \in \mathbb{R}^d$, $k \ll d$, such that $S = \operatorname{span}\{u_1, \dots, u_k\}$ captures as much variation as possible. Then

$$\underbrace{x \in \mathbb{R}^d}_{\text{original}} \longmapsto \underbrace{z = \begin{pmatrix} u_1^\top x \\ \vdots \\ u_k^\top x \end{pmatrix} \in \mathbb{R}^k}_{\text{compressed code}} \longmapsto \underbrace{\hat{x} = \sum_{j=1}^{k} (u_j^\top x) u_j \in S}_{\text{reconstruction}}.$$

A face image with $d = 10\,000$ pixels can be summarised by $k = 100$ PCA coordinates.



TL;DR // Rank–nullity = information accounting

$\operatorname{rank} L$: number of independent input directions **preserved**. $\dim \ker L$: number of independent input directions **collapsed to zero**. They add up to $\dim U$. You can't compress without losing *something* unless the data already lived in a smaller subspace.

7 The exercises, fully solved

TL;DR // How to use this section

Try each question first, then check. Each solution is written the way you'd write it in an exam: introduce your variables, justify every step, conclude with a sentence.

7.1 Part 1(a): Theory questions

Question 1 – Euclidean space

A data point is $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$.

SOLUTION // Q1(a) Why is $\mathbb{R}^d$ a vector space? Which operations?

The operations are component-wise addition and scalar multiplication: for $x, y \in \mathbb{R}^d$, $\lambda \in \mathbb{R}$, $x + y = (x_1 + y_1, \ldots, x_d + y_d)$ and $\lambda x = (\lambda x_1, \ldots, \lambda x_d)$. Both land in $\mathbb{R}^d$ (closure). Each axiom reduces, coordinate by coordinate, to a property of real numbers: associativity and commutativity of $+$ in $\mathbb{R}$, zero vector $0 = (0, \ldots, 0)$, opposite $-x = (-x_1, \ldots, -x_d)$, distributivity $\lambda(x_j + y_j) = \lambda x_j + \lambda y_j$, $(\alpha + \beta)x_j = \alpha x_j + \beta x_j$, $\alpha(\beta x_j) = (\alpha\beta)x_j$, $1 \cdot x_j = x_j$. Equivalently: $\mathbb{R}^d = \mathbb{R} \times \cdots \times \mathbb{R}$ is a product of d copies of the vector space $\mathbb{R}$, and a product of vector spaces is a vector space.

SOLUTION // Q1(b) $S = \{x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 0\}$ is a subspace

Zero: $\sum_{j=1}^d 0 = 0$, so $0 \in S$. *Stability:* let $x, y \in S$ and $\lambda \in \mathbb{R}$. Then $\sum_{j=1}^d (\lambda x_j + y_j) = \lambda \sum_{j=1}^d x_j + \sum_{j=1}^d y_j = \lambda \cdot 0 + 0 = 0$, so $\lambda x + y \in S$ (this covers both $x + y$, with $\lambda = 1$, and λx, with $y = 0$). Hence S is a subspace. (It's the homogeneous-equation proposition with $a_1 = \cdots = a_d = 1$; also $S = \ker(x \mapsto \sum_j x_j)$.) Its dimension is $d - 1$: a hyperplane.
ML meaning: S is where **centred** features live when you centre across features, or where vectors "summing to zero" live (e.g. differences of probability vectors).

SOLUTION // Q1(c) A subset that is not a subspace

Three classic ones (any one suffices, with the failing axiom):

1. $A = \{x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 1\}$ (e.g. probability vectors): $0 \notin A$ since $\sum_j 0 = 0 \neq 1$. Axiom (i) fails. (Also not closed under $+$: sum of two elements has coordinates summing to 2.)
2. $\mathbb{R}^d_+ = \{x \mid x_j \geq 0 \ \forall j\}$ (e.g. pixel intensities): contains 0 and is closed under $+$, but $x = (1, 0, \ldots, 0)$ gives $-x \notin \mathbb{R}^d_+$. Scalar multiplication fails.
3. $\mathrm{span}(e_1) \cup \mathrm{span}(e_2)$: contains 0, closed under scaling, but $e_1 + e_2 \notin$ it. Addition fails.

Question 2 – Span (dictionary learning)

SOLUTION // Q2(a) Definition

$\mathrm{span}(d_1, \ldots, d_k) = \left\{ \sum_{i=1}^k c_i d_i \;\middle|\; c_1, \ldots, c_k \in \mathbb{R} \right\}$: the set of all linear combinations of the atoms.

SOLUTION // Q2(b) Why a span is always a subspace

$0 = \sum_{i=1}^k 0 \cdot d_i \in$ span. If $x = \sum_{i=1}^k c_i d_i$ and $y = \sum_{i=1}^k c'_i d_i$, then for $\lambda \in \mathbb{R}$, $\lambda x + y = \sum_{i=1}^k (\lambda c_i + c'_i) d_i$, again a linear combination of the atoms. So it contains 0 and is stable under $+$ and scaling: a subspace.

SOLUTION // Q2(c) Geometric meaning of $x \in \mathrm{span}(d_1, \ldots, d_k)$

The span is a "flat" through the origin (a line if it has dimension 1, a plane if 2, a hyperplane…). $x \in$ span means x lies *exactly* on this flat, i.e. x can be **perfectly reconstructed** as $x =$

$c_1 d_1 + \cdots + c_k d_k$ with zero error. If $x \notin$ span, the best we can do is approximate x by the closest point of the flat (the projection), with a non-zero residual.

SOLUTION // Q2(d) What goes wrong if the atoms are linearly dependent?

1. **Non-unique coefficients.** If $\sum_i a_i d_i = 0$ with $a \neq 0$, then $x = \sum_i c_i d_i = \sum_i (c_i + t a_i) d_i$ for every $t \in \mathbb{R}$: infinitely many codes for the same x. The coefficients c are no longer interpretable ("how much of pattern i" is ambiguous).
2. **Redundancy.** $\dim \operatorname{span}(d_1, \ldots, d_k) < k$: you pay for k atoms but only get fewer directions. At least one atom is useless.
3. **Numerical instability.** The system $Dc \approx x$ is ill-posed (D is singular); least-squares solutions blow up, small noise $\Rightarrow$ big changes in c.

(Side note: in real dictionary learning, *overcomplete* dictionaries with $k > n$ are used on purpose, and a sparsity penalty picks one of the many representations. But in linear-algebra terms, the representation is then not unique.)

Question 3 – Basis, dimension and embeddings

SOLUTION // Q3(a) One-hot vectors form a basis of $\mathbb{R}^K$

Spanning: any $x = (x_1, \ldots, x_K) = \sum_{j=1}^K x_j e_j$. *Independent:* $\sum_{j=1}^K c_j e_j = (c_1, \ldots, c_K) = 0 \Rightarrow c_1 = \cdots = c_K = 0$. So $(e_1, \ldots, e_K)$ is a basis (the canonical one) and $\dim \mathbb{R}^K = K$.

SOLUTION // Q3(b) One-hot in $\mathbb{R}^K$ vs embeddings in $\mathbb{R}^r$, $r \ll K$

1. **Dimension drops** from K to r: vectors are short and dense instead of long and sparse (memory and compute win).
2. **No more independence.** In $\mathbb{R}^r$ at most r vectors can be linearly independent, so with $K > r$ categories the embeddings $x_1, \ldots, x_K$ are necessarily *linearly dependent*: they are not a basis anymore. Categories share directions.
3. **Geometry now carries meaning.** One-hot vectors are pairwise orthogonal and all at distance $\sqrt{2}$: no notion of similarity. Embeddings are learned so that similar categories are close ("cat" near "dog").
4. **It's a linear map.** An embedding table is a matrix $M \in \mathbb{R}^{r \times K}$ and the embedding of category j is Me_j (the j-th column): a linear map $\mathbb{R}^K \to \mathbb{R}^r$. Since $K > r$ it can't be injective (rank $\leq r < K$, so $\dim \ker M \geq K - r > 0$), though the K specific vectors Me_j can still be distinct.

Question 4 – Linear maps, kernel, range, rank

SOLUTION // Q4(a)(b) Definitions and rank–nullity

For $L : U \to V$ linear: $\ker L = \{x \in U \mid L(x) = 0_V\}$, $\operatorname{range} L = \{L(x) \mid x \in U\}$, $\operatorname{rank} L = \dim \operatorname{range} L$. Rank–nullity (for U finite-dimensional): $\operatorname{rank} L + \dim \ker L = \dim U$.

SOLUTION // Q4(c) $x \mapsto Wx$ is linear

Let $W \in \mathbb{R}^{k\times d}$, $x, y \in \mathbb{R}^d$, $a, b \in \mathbb{R}$. By distributivity of matrix multiplication and compatibility with scalars, $W(ax + by) = aWx + bWy$. Coordinate-wise: $(W(ax + by))_i = \sum_{j=1}^d W_{ij}(ax_j + by_j) = a\sum_{j=1}^d W_{ij}x_j + b\sum_{j=1}^d W_{ij}y_j$. So it's linear.

SOLUTION // Q4(d) Why $x \mapsto Wx + b$ is not linear (in general)

Any linear map sends 0 to 0. Here $f(0) = W0 + b = b$. If $b \neq 0$, $f(0) \neq 0$, so f is not linear. (Also $f(2x) = 2Wx + b \neq 2Wx + 2b = 2f(x)$ when $b \neq 0$.) It is **affine**: linear map + constant shift. Linear only when $b = 0$.

7.2 Part 1(b): True or False

SOLUTION // T/F 1 – "Every subset of $\mathbb{R}^d$ containing 0 is a subspace"

FALSE. Counterexample in $\mathbb{R}^2$: $\{(0,0),(1,0)\}$ contains 0 but $2(1,0) = (2,0) \notin$ it. Or the union of the two axes (not closed under +). Containing 0 is necessary, not sufficient.

SOLUTION // T/F 2 – "Non-empty + closed under + and scaling ⇒ subspace"

TRUE. Take any $x \in S$ (exists since non-empty). By stability under scaling, $0 \cdot x = 0 \in S$. So all three conditions hold. (That's why "non-empty" can replace "$0 \in S$" in the criterion.)

SOLUTION // T/F 3 – "$A = \{x \mid \sum_{j=1}^d x_j = 1\}$ is a subspace"

FALSE. $0 \notin A$ since $\sum_j 0 = 0 \neq 1$. It's an *affine* hyperplane (a subspace translated away from the origin).

SOLUTION // T/F 4 – "The span of a finite family is always a subspace"

TRUE. Proved in Q2(b). (Convention: the span of the empty family is $\{0\}$, also a subspace.)

SOLUTION // T/F 5 – "$v_1, \ldots, v_k$ independent in $\mathbb{R}^d \Rightarrow k \leq d$"

TRUE. $(e_1, \ldots, e_d)$ spans $\mathbb{R}^d$ with d vectors; by the exchange lemma, an independent family has at most d vectors.

SOLUTION // T/F 6 – "$v_1, \ldots, v_d$ span $\mathbb{R}^d \Rightarrow$ basis"

TRUE. Extract a basis from the spanning family; it must have exactly $\dim \mathbb{R}^d = d$ elements, so we removed nothing: the family itself is a basis (in particular independent). "n vectors in dimension n: spanning $\iff$ independent."

SOLUTION // T/F 7 – "$L : \mathbb{R}^d \to \mathbb{R}^k$ linear with $k < d$ cannot be injective"

TRUE. $\operatorname{rank} L \leq k$ (range $\subseteq \mathbb{R}^k$), so $\dim \ker L = d - \operatorname{rank} L \geq d - k > 0$: the kernel contains non-zero vectors, so L isn't injective. *You can't squash d dimensions into fewer without collisions.*

SOLUTION // T/F 8 – "$\ker L = \{0\} \Rightarrow L$ injective"

TRUE (for L linear). If $L(x) = L(y)$ then $L(x - y) = 0$, so $x - y \in \ker L = \{0\}$, $x = y$. (For non-linear functions this is false: $x \mapsto x^2$ only sends 0 to 0 but isn't injective.)

SOLUTION // T/F 9 – "$L : \mathbb{R}^d \to \mathbb{R}^k$ with $\operatorname{rank} L = k$ is surjective"

TRUE. $\operatorname{range} L$ is a subspace of $\mathbb{R}^k$ with dimension $k = \dim \mathbb{R}^k$, so $\operatorname{range} L = \mathbb{R}^k$.

SOLUTION // T/F 10 – "A feature that is an exact linear combination of others ⇒ dependent columns"

TRUE. If column $c_j = \sum_{i \neq j} a_i c_i$, then $\sum_{i \neq j} a_i c_i - 1 \cdot c_j = 0$ is a non-trivial combination (coefficient $-1 \neq 0$ on c_j). E.g. height in m and in cm: $100c_1 - c_2 = 0$.

SOLUTION // T/F 11 – "PCA coordinates = linear combinations of basis directions"

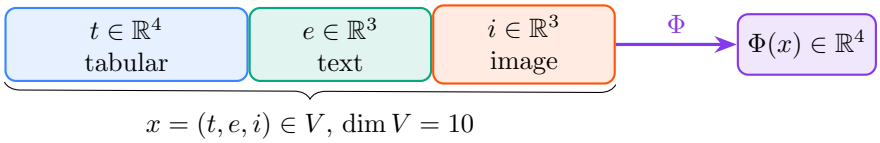
TRUE. The PCA approximation is $\hat{x} = \sum_{j=1}^{k} z_j u_j$ with $z_j = u_j^\top x$: a linear combination of the basis $(u_1, \ldots, u_k)$ of the subspace S; the coordinates z_j are the representation.

7.3 Part 2: The product-representation exercise

Setup. Products are represented by $x = (t, e, i) \in V = \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3$ with tabular features t, text embedding e, image embedding i. Admissible representations satisfy

$$t_2 = 100t_1, \qquad t_4 = t_1 + t_3, \qquad e_1 + e_2 + e_3 = 0, \qquad i \in \operatorname{span}(a, b),\ a = (1, 0, 1),\ b = (0, 1, 1).$$

$S \subseteq V$ is the set of admissible representations.



$t_2 = 100t_1,\ t_4 = t_1 + t_3 \quad \sum e_j = 0 \quad i \in \operatorname{span}(a, b)$

SOLUTION // 1. Dimension of V

By the product formula: $\dim V = \dim \mathbb{R}^4 + \dim \mathbb{R}^3 + \dim \mathbb{R}^3 = 4 + 3 + 3 = \boxed{10}$.

SOLUTION // 2. S is a subspace of V

Note $S = S_t \times S_e \times S_i$ where S_t, S_e, S_i are defined in question 3. Direct check: *Zero:* $x = (0, 0, 0)$ satisfies $0 = 100 \cdot 0$, $0 = 0 + 0$, $0 + 0 + 0 = 0$, and $0 = 0a + 0b \in \operatorname{span}(a, b)$. So $0_V \in S$. *Stability:* let $x = (t, e, i)$, $x' = (t', e', i') \in S$ and $\lambda \in \mathbb{R}$; set $y = \lambda x + x' = (\lambda t + t', \lambda e + e', \lambda i + i')$.

- $\lambda t_2 + t'_2 = \lambda(100t_1) + 100t'_1 = 100(\lambda t_1 + t'_1)$ ✓
- $\lambda t_4 + t'_4 = \lambda(t_1 + t_3) + (t'_1 + t'_3) = (\lambda t_1 + t'_1) + (\lambda t_3 + t'_3)$ ✓
- $\sum_{j=1}^{3} (\lambda e_j + e'_j) = \lambda \cdot 0 + 0 = 0$ ✓
- $i = \alpha a + \beta b$, $i' = \alpha' a + \beta' b \Rightarrow \lambda i + i' = (\lambda\alpha + \alpha')a + (\lambda\beta + \beta')b \in \operatorname{span}(a, b)$ ✓

So $y \in S$. Hence S is a subspace of V. (Slicker: all constraints are homogeneous linear equations or a span, and a product of subspaces is a subspace.)

SOLUTION // 3. Bases and dimensions of S_t, S_e, S_i

S_t. Free variables t_1, t_3; then $t_2 = 100t_1$, $t_4 = t_1 + t_3$:

$$t = (t_1, 100t_1, t_3, t_1 + t_3) = t_1 \underbrace{(1, 100, 0, 1)}_{f_1} + t_3 \underbrace{(0, 0, 1, 1)}_{f_2}.$$

$S_t = \operatorname{span}(f_1, f_2)$. Independent: $\alpha f_1 + \beta f_2 = (\alpha, 100\alpha, \beta, \alpha + \beta) = 0$ forces $\alpha = \beta = 0$ (look at coordinates 1 and 3). Basis (f_1, f_2), $\dim S_t = 2$ $(= 4 - 2$ equations$)$.
S_e. $e_3 = -e_1 - e_2$: $e = (e_1, e_2, -e_1 - e_2) = e_1(1, 0, -1) + e_2(0, 1, -1)$. Basis $g_1 = (1, 0, -1)$, $g_2 = (0, 1, -1)$ (independent: first two coordinates), $\dim S_e = 2$.
S_i. $S_i = \operatorname{span}(a, b)$ spans by definition. Independent: $\alpha(1, 0, 1) + \beta(0, 1, 1) = (\alpha, \beta, \alpha + \beta) = 0 \Rightarrow \alpha = \beta = 0$. Basis (a, b), $\dim S_i = 2$. (Equation form: $i \in S_i \iff i_3 = i_1 + i_2$.)
Bonus: $\dim S = \dim S_t + \dim S_e + \dim S_i = 2 + 2 + 2 = 6$. The constraints kill 4 of the 10 dimensions.

Now the model keeps only $\Phi(x) = \big(t_1,\ t_3,\ e_1 - e_2,\ i_1 + i_2 + i_3\big)^\top \in \mathbb{R}^4$.

SOLUTION // 4. $\Phi : S \to \mathbb{R}^4$ is linear

Each component is a linear form in the coordinates of x (no constants, no products). Explicitly, for $x, x' \in S$, $\lambda, \mu \in \mathbb{R}$:

$$\Phi(\lambda x + \mu x') = \begin{pmatrix} \lambda t_1 + \mu t_1' \\ \lambda t_3 + \mu t_3' \\ (\lambda e_1 + \mu e_1') - (\lambda e_2 + \mu e_2') \\ \sum_{j=1}^{3} (\lambda i_j + \mu i_j') \end{pmatrix} = \lambda \begin{pmatrix} t_1 \\ t_3 \\ e_1 - e_2 \\ \sum_j i_j \end{pmatrix} + \mu \begin{pmatrix} t_1' \\ t_3' \\ e_1' - e_2' \\ \sum_j i_j' \end{pmatrix} = \lambda\Phi(x) + \mu\Phi(x').$$

(And S is a vector space by Q2, so it makes sense to talk about a linear map on S.)

SOLUTION // 5. Is Φ injective?

No. Dimension argument: $\dim S = 6 > 4 = \dim \mathbb{R}^4$, so $\dim \ker \Phi = 6 - \operatorname{rank} \Phi \geq 6 - 4 = 2 > 0$. Explicit collision: $x = (0,\ (1, 1, -2),\ 0) \in S$ (since $1 + 1 - 2 = 0$) is non-zero but $\Phi(x) = (0, 0, 1 - 1, 0) = 0 = \Phi(0)$.

SOLUTION // 6. $\ker \Phi$ and its dimension

Parametrise S with the bases from Q3: $t = t_1 f_1 + t_3 f_2$, $e = (e_1, e_2, -e_1 - e_2)$, $i = \alpha a + \beta b = (\alpha, \beta, \alpha + \beta)$. Then

$$\Phi(x) = \big(t_1,\ t_3,\ e_1 - e_2,\ \alpha + \beta + (\alpha + \beta)\big) = \big(t_1,\ t_3,\ e_1 - e_2,\ 2(\alpha + \beta)\big).$$

$\Phi(x) = 0 \iff t_1 = 0,\ t_3 = 0,\ e_1 = e_2,\ \beta = -\alpha$. Then $t = 0$, $e = (e_1, e_1, -2e_1) = e_1(1, 1, -2)$, $i = \alpha(a - b) = \alpha(1, -1, 0)$. So

$$\ker \Phi = \operatorname{span}\Big(\underbrace{(0_{\mathbb{R}^4}, (1, 1, -2), 0_{\mathbb{R}^3})}_{k_1},\ \underbrace{(0_{\mathbb{R}^4}, 0_{\mathbb{R}^3}, (1, -1, 0))}_{k_2} \Big).$$

k_1, k_2 are independent (they live in different blocks), so $\dim \ker \Phi = \boxed{2}$.

SOLUTION // 7. $\operatorname{rank} \Phi$ by rank–nullity

$\operatorname{rank} \Phi = \dim S - \dim \ker \Phi = 6 - 2 = \boxed{4}$.

SOLUTION // 8. Is Φ surjective onto $\mathbb{R}^4$?

Yes: $\operatorname{rank}\Phi = 4 = \dim\mathbb{R}^4$. Explicitly, for $(y_1, y_2, y_3, y_4) \in \mathbb{R}^4$ take $t_1 = y_1$, $t_3 = y_2$, $e = (y_3, 0, -y_3)$, $\alpha = \beta = y_4/4$: then $\Phi(x) = (y_1, y_2, y_3 - 0,\ 2 \cdot \frac{y_4}{2}) = (y_1, y_2, y_3, y_4)$ ✓.

SOLUTION // 9. Interpretation of $\ker\Phi$ (what info is lost)

Two admissible products x, x' with $x - x' \in \ker\Phi$ look *identical* to the model. Concretely:

- **Tabular: nothing lost.** The kernel has no t-component: t_2 and t_4 are determined by t_1, t_3 anyway (redundant features), so keeping t_1, t_3 is lossless.
- **Text: 1 direction lost**, $(1, 1, -2)$. Φ only keeps $e_1 - e_2$; within S_e (2D), the direction where $e_1 = e_2$ (e.g. how $e_1 + e_2 = -e_3$ compares to e_3) is invisible.
- **Image: 1 direction lost**, $(1, -1, 0) = a - b$. Φ only keeps $i_1 + i_2 + i_3 = 2(\alpha + \beta)$, so it can't see how the image splits between the two "patterns" a and b (it sees $\alpha + \beta$ but not $\alpha - \beta$).

Rank–nullity accounting: of the 6 admissible directions, 4 are kept and 2 are discarded.

Prediction model: $g(x) = w^\top\Phi(x) + \beta$, $w \in \mathbb{R}^4$, $\beta \in \mathbb{R}$.

SOLUTION // 10. $x \mapsto w^\top\Phi(x)$ is linear on S

It's the composition of $\Phi : S \to \mathbb{R}^4$ (linear) and $y \mapsto w^\top y = \sum_{j=1}^4 w_j y_j$ (linear). Directly: $w^\top\Phi(\lambda x + \mu x') = w^\top(\lambda\Phi(x) + \mu\Phi(x')) = \lambda w^\top\Phi(x) + \mu w^\top\Phi(x')$.

SOLUTION // 11. g is not linear when $\beta \neq 0$

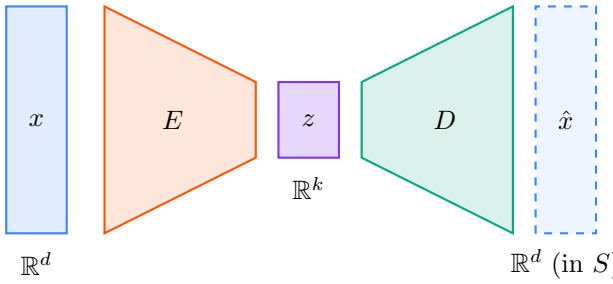
$g(0) = w^\top\Phi(0) + \beta = w^\top 0 + \beta = \beta \neq 0$, but a linear map must send 0 to 0. So g is not linear (it's affine).

SOLUTION // 12. Why bias terms are useful

Without bias, every prediction function passes through the origin: $g(0) = 0$ is forced. With bias the model can learn an **offset**: a baseline value when all features are 0 (e.g. a base price), a decision boundary $w^\top\Phi(x) + \beta = 0$ that doesn't have to go through the origin, and it can absorb non-zero means of the data/targets. Geometrically: linear maps give subspaces, affine maps give *shifted* ones. (Trick: append a constant feature 1 to make an affine map linear in one extra dimension, $w^\top\Phi(x) + \beta = (w, \beta)^\top(\Phi(x), 1)$.)

7.4 Part 3: A bit of ML – linear encoder/decoder

Setup. $x \in \mathbb{R}^d$ is an image, $u_1, \ldots, u_k \in \mathbb{R}^d$ are orthonormal ($u_i^\top u_j = 1$ if $i = j$, 0 otherwise). Encoder $E : \mathbb{R}^d \to \mathbb{R}^k$, $E(x) = (u_1^\top x, \ldots, u_k^\top x)^\top$; decoder $D : \mathbb{R}^k \to \mathbb{R}^d$, $D(z) = z_1 u_1 + \cdots + z_k u_k$; reconstruction $\hat{x} = D(E(x))$.



encoder: d numbers $\to$ k numbers
bottleneck z = the code / embedding
decoder: rebuild in $S = \text{span}(u_1, \ldots, u_k)$
$\hat{x} = \sum_{j=1}^{k} (u_j^\top x)\, u_j$

Step 1 – Subspaces and dimension

SOLUTION // 1. Why images are in $\mathbb{R}^d$

An image is d real pixel intensities $x = (x_1, \ldots, x_d)$, i.e. an element of $\mathbb{R}^d$, which is a vector space (Q1(a)). Adding images and scaling them are pixel-wise operations.

SOLUTION // 2. $S = \text{span}(u_1, \ldots, u_k)$ is a subspace

A span is always a subspace (Q2(b)): contains 0, stable under + and scaling.

SOLUTION // 3. $\dim S = k$

$(u_1, \ldots, u_k)$ spans S by definition; it remains to show independence. Suppose $\sum_{i=1}^{k} c_i u_i = 0$. Take the dot product with u_j (fixed j):

$$0 = u_j^\top \Big(\sum_{i=1}^{k} c_i u_i \Big) = \sum_{i=1}^{k} c_i\, u_j^\top u_i = c_j \cdot 1 + \sum_{i \neq j} c_i \cdot 0 = c_j.$$

So every $c_j = 0$: independent, hence a basis of S, and $\dim S = k$. *Orthonormal $\Rightarrow$ independent. Always. Dot-product trick.*

SOLUTION // 4. Why this is dimensionality reduction (geometry)

S is a k-dimensional flat through 0 inside $\mathbb{R}^d$ (a line, a plane…). Once we replace x by a point of S, that point is fully described by its k coordinates in the basis $(u_1, \ldots, u_k)$ instead of d numbers. We moved from a d-dim world to a k-dim one, $k \ll d$.

Step 2 – Encoding

SOLUTION // 5. E is linear

Each coordinate $x \mapsto u_j^\top x = \sum_{l=1}^{d} u_{j,l} x_l$ is linear: $u_j^\top (ax + by) = a\, u_j^\top x + b\, u_j^\top y$. Stacking linear coordinates gives a linear map: $E(ax + by) = aE(x) + bE(y)$. (Matrix form: $E(x) = U^\top x$ with $U = [u_1 \ \cdots \ u_k] \in \mathbb{R}^{d \times k}$.)

SOLUTION // 6. Range of E; is it surjective?

Let $z \in \mathbb{R}^k$. Try $x = D(z) = \sum_{l=1}^{k} z_l u_l$. Then $E(x)_j = u_j^\top \sum_{l=1}^{k} z_l u_l = \sum_{l=1}^{k} z_l\, u_j^\top u_l = z_j$. So $E(D(z)) = z$: every z is reached. range $E = \mathbb{R}^k$, E is **surjective**, rank $E = k$.

SOLUTION // 7. $\dim\ker E$

Rank–nullity: $\dim\ker E = d - \operatorname{rank} E = \boxed{d-k}$.

SOLUTION // 8. Describe $\ker E$

$E(x) = 0 \iff u_1^\top x = 0,\ \ldots,\ u_k^\top x = 0$. So $\ker E = \{x \in \mathbb{R}^d \mid u_j^\top x = 0 \text{ for all } j = 1,\ldots,k\}$: the vectors **orthogonal to every** u_j, i.e. orthogonal to all of S (the orthogonal complement $S^\perp$, more on this in later lectures).

SOLUTION // 9. Interpretation of $\ker E$

$\ker E$ is what the encoder is **blind** to: if $x' = x + n$ with $n \in \ker E$, then $E(x') = E(x)$. Those $d-k$ directions of pixel space are thrown away (in PCA: the low-variance directions, often noise or fine detail).

Step 3 – Decoding

SOLUTION // 10. D is linear

$D(az+bz') = \sum_{j=1}^k (az_j + bz'_j)u_j = a\sum_{j=1}^k z_j u_j + b\sum_{j=1}^k z'_j u_j = aD(z) + bD(z')$. (Matrix form: $D(z) = Uz$.)

SOLUTION // 11. $\operatorname{range} D$

$\operatorname{range} D = \{z_1u_1 + \cdots + z_ku_k \mid z \in \mathbb{R}^k\} = \operatorname{span}(u_1,\ldots,u_k) = S$.

SOLUTION // 12. Is D injective?

Yes. If $D(z) = \sum_{j=1}^k z_j u_j = 0$, independence of the u_j (Step 1, Q3) gives $z = 0$. So $\ker D = \{0\}$. (Equivalently: $\operatorname{rank} D = \dim S = k = \dim \mathbb{R}^k$.)

SOLUTION // 13. Is D surjective onto $\mathbb{R}^d$ when $k < d$?

No. $\operatorname{rank} D = k < d = \dim \mathbb{R}^d$, so $\operatorname{range} D = S \neq \mathbb{R}^d$. Concretely: any non-zero $n \in \ker E$ (exists since $\dim\ker E = d - k > 0$) satisfies $u_j^\top n = 0$ for all j, and if $n = D(z)$ then $z = E(D(z)) = E(n) = 0$, so $n = 0$, contradiction. The decoder can only produce images inside S.

Step 4 – Reconstruction

SOLUTION // 14–15. Explicit formula for $\hat{x}$

$z = E(x)$ has coordinates $z_j = u_j^\top x$. Plugging into D:

$$\hat{x} = D(E(x)) = z_1u_1 + \cdots + z_ku_k = \sum_{j=1}^k (u_j^\top x)\, u_j.$$

(Matrix form: $\hat{x} = UU^\top x$.)

SOLUTION // 16. $\hat{x} \in S$ for every x

$\hat{x}$ is a linear combination of $u_1, \dots, u_k$ (coefficients $u_j^\top x \in \mathbb{R}$), hence in $\operatorname{span}(u_1, \dots, u_k) = S$. (Or: $\hat{x} \in \operatorname{range} D = S$.)

SOLUTION // 17. If $x \in S$ then $\hat{x} = x$

Write $x = \sum_{i=1}^k c_i u_i$. Then $u_j^\top x = \sum_{i=1}^k c_i\, u_j^\top u_i = c_j$. So $\hat{x} = \sum_{j=1}^k c_j u_j = x$. *Things already in S survive compression perfectly.*

SOLUTION // 18. $\hat{x}$ = the part of x that is kept

Write $x = \hat{x} + r$ with $r = x - \hat{x}$. Then $\hat{x} \in S$, and for every j: $u_j^\top r = u_j^\top x - u_j^\top \hat{x} = u_j^\top x - u_j^\top x = 0$ (by 17 applied coordinate-wise, $u_j^\top \hat{x} = u_j^\top x$). So $r \in \ker E$. Hence x splits into a piece in S (kept: E only sees it, D rebuilds it) plus a piece in $\ker E$ (invisible to the encoder, lost). $\hat{x}$ is exactly the kept piece: the **projection** of x onto S. Moreover $S \cap \ker E = \{0\}$ (if $x \in S$ and $E(x) = 0$ then $x = \hat{x} = D(0) = 0$), and $\dim S + \dim \ker E = k + (d - k) = d$, so $\mathbb{R}^d = S \oplus \ker E$.

Step 5 [Bonus] – Concrete example, $d = 5$, $k = 2$

$u_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0, 0)$, $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1, 0)$.

SOLUTION // 19. Orthonormality

$u_1^\top u_1 = \frac{1}{2}(1^2 + 1^2) = 1$, $u_2^\top u_2 = \frac{1}{2}(1 + 1) = 1$, $u_1^\top u_2 = \frac{1}{2}(1 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 0) = 0$. ✓

SOLUTION // 20. $E(x)$ for $x = (2, 4, 1, 3, 10)$

$u_1^\top x = \frac{2+4}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$, $u_2^\top x = \frac{1+3}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. So $\boxed{E(x) = (3\sqrt{2},\ 2\sqrt{2})} \approx (4.24,\ 2.83)$.

SOLUTION // 21. $\hat{x} = D(E(x))$

$\hat{x} = 3\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1, 0, 0, 0) + 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(0, 0, 1, 1, 0) = (3, 3, 0, 0, 0) + (0, 0, 2, 2, 0) = \boxed{(3, 3, 2, 2, 0)}$. *The encoder replaced pixels 1–2 by their average (3), pixels 3–4 by their average (2), and deleted pixel 5.*

SOLUTION // 22. What is completely lost

Residual $x - \hat{x} = (2 - 3,\ 4 - 3,\ 1 - 2,\ 3 - 2,\ 10 - 0) = (-1, 1, -1, 1, 10)$. Lost: the **difference** between pixels 1 and 2 ($x_1 - x_2$), the difference between pixels 3 and 4 ($x_3 - x_4$), and **pixel 5 entirely** (its value 10, the brightest pixel, is gone!).

SOLUTION // 23. $\ker E$ explicitly

$E(x) = 0 \iff x_1 + x_2 = 0$ and $x_3 + x_4 = 0$ (x_5 free). So $x = (x_1, -x_1, x_3, -x_3, x_5)$:

$$\ker E = \operatorname{span}\big((1, -1, 0, 0, 0),\ (0, 0, 1, -1, 0),\ (0, 0, 0, 0, 1)\big), \qquad \dim \ker E = 3.$$

(Check: the residual $(-1, 1, -1, 1, 10) = -1 \cdot (1, -1, 0, 0, 0) - 1 \cdot (0, 0, 1, -1, 0) + 10 \cdot (0, 0, 0, 0, 1) \in \ker E$ ✓.)

SOLUTION // 24. Rank–nullity check

$\operatorname{rank} E = 2$ (E is surjective onto $\mathbb{R}^2$), $\dim \ker E = 3$, and $2 + 3 = 5 = d$ ✓.

Step 6 [Bonus] – Link with PCA

SOLUTION // 25–28. The PCA dictionary

- Low-dimensional subspace: $S = \operatorname{span}(u_1, \dots, u_k)$.
- Compressed representation: $z = E(x) = (u_1^\top x, \dots, u_k^\top x) \in \mathbb{R}^k$.
- Reconstructed approximation: $\hat{x} = D(E(x)) = \sum_{j=1}^{k} (u_j^\top x) u_j \in S$.
- Why "linear dimensionality reduction": E, D and $D \circ E$ are *linear* maps, and the code has dimension $k < d$. PCA's only extra ingredient: choose the u_j from the data to keep the most variance.

SOLUTION // 29. $d = 10\,000$, $k = 100$

We keep 100 coordinates, i.e. $\frac{100}{10\,000} = \frac{1}{100} = \boxed{1\%}$ of the original. 99% compression.

SOLUTION // 30. Why low-dim representations help retrieval / face recognition

- **Speed & memory:** comparing two faces costs $k = 100$ operations instead of $10\,000$; storing a database is $100\times$ cheaper.
- **Denoising:** discarded directions (low variance) are often noise, lighting glitches, irrelevant pixel detail; the kept ones capture the main variation (identity, pose…).
- **Better geometry:** in raw pixel space, distances are dominated by irrelevant pixel differences (curse of dimensionality); in the compressed space, nearby codes tend to mean similar faces (the "eigenfaces" idea).

Step 7 [Bonus] – Deep image embeddings

SOLUTION // 31. Retrieval with embeddings

Compute $z_q = E(x_q)$. For each database image compute the distance $\|z_q - z^{(i)}\|_2$ (or cosine similarity $\frac{z_q^\top z^{(i)}}{\|z_q\| \|z^{(i)}\|}$). Return the images with the smallest distances (the K nearest neighbours). Retrieval = nearest-neighbour search in $\mathbb{R}^k$ (approximate NN indexes make it fast at scale).

SOLUTION // 32. Linear encoder vs neural encoder

E here is **linear**: $E(ax + by) = aE(x) + bE(y)$, it's a single matrix $U^\top$. A neural encoder is a composition of affine maps and **non-linear** activations, $x \mapsto \sigma(W_L \cdots \sigma(W_1 x + b_1) \cdots + b_L)$: it's not linear, so no kernel/range/rank-nullity in the strict sense, and it can "bend" the data (capture curved structures a subspace can't).

SOLUTION // 33. Why study the linear case first

- Neural networks are *built* from linear maps (Wx) + biases + simple non-linearities: you need the linear part cold.

- The embedding space $\mathbb{R}^k$ is still a vector space: distances, averages, spans, subspaces all still make sense there.
- Locally, a smooth non-linear encoder behaves like a linear map (its Jacobian): kernel/rank ideas describe which input directions it locally ignores.
- PCA is the go-to baseline and it's exactly this linear encoder; it has closed-form solutions and is fully interpretable.

8 Final boss: the one-page cheat sheet

8.1 "If they ask you this, do that"

If asked...	Do this
Show S is a subspace	$0 \in S$; take $x, y \in S$, $\lambda \in \mathbb{R}$; show $\lambda x + y \in S$. Or: write S as a span / kernel / intersection.
Show S is NOT a subspace	Check $0 \notin S$ first. Else explicit x with $-x \notin S$ or x, y with $x + y \notin S$.
Find a basis / dim of $S = \{\text{eqs}\}$	Solve for pivots, write $x = \sum(\text{free var}) \cdot v_i$, check independence, dim = #free vars.
Show a family spans V	Arbitrary x, solve $\sum \lambda_i v_i = x$ for the λ_i.
Show a family is independent	$\sum \alpha_i v_i = 0 \Rightarrow$ solve $\Rightarrow$ all $\alpha_i = 0$. If dependent, exhibit the non-zero α.
Show it's a basis	Independent + spanning; if #vectors = dim, only one of the two.
Show L linear	Compute $L(au + bu')$ and $aL(u) + bL(u')$, show equal.
Show L NOT linear	$L(0) \neq 0$, or explicit counterexample.
Find $\ker L$	Solve $L(x) = 0$, parametrise, basis, dim.
Find $\operatorname{range} L$ / $\operatorname{rank} L$	span of images of a basis, extract a basis; or rank = $\dim U - \dim \ker$.
Injective?	Linear: $\ker L = \{0\}$? Quick NO if $\dim U > \dim V$.
Surjective?	$\operatorname{rank} L = \dim V$? Quick NO if $\dim U < \dim V$. Or solve $L(x) = y$ for arbitrary y.
Bijective?	Both; if $\dim U = \dim V$ just one of them.
Use rank–nullity	$\operatorname{rank} L + \dim \ker L = \dim(\text{DOMAIN})$. Never the codomain!
dim of a sum	$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.
Interpret kernel (ML)	Directions/information the model can't see: inputs differing by a kernel vector get the same output.

8.2 Hall of Ls (classic mistakes)

COMMON L (don't do this) // Do not

1. Say "S contains 0, so it's a subspace." Necessary, not sufficient.
2. Use rank–nullity with the dimension of the **codomain**. It's dim(domain).
3. Say "surjective" without specifying onto what. The codomain matters.
4. Confuse $f^{-1}(y)$ (preimage set, always defined) with the inverse function (only for bijections).
5. Think "two vectors $\Rightarrow$ span is 2D". Only if independent (span$((1,0),(3,0))$ is a line).

6. Call $x \mapsto Wx + b$ linear. It's affine (unless $b = 0$).
7. Prove independence by "they look different". Solve the system.
8. Forget that for non-linear functions, $f^{-1}(0) = \{0\}$ does NOT imply injective (x^2).
9. Write a sum without its range, or use a letter you never introduced.

8.3 The whole lecture in 10 lines

TL;DR // Screenshot this

1. **Vector space**: add + scale, 8 axioms. $\mathbb{R}^d$, matrices, polynomials, functions.
2. **Subspace**: $0 \in S$, closed under + and scaling. Lines/planes *through* 0.
3. **Span**: all linear combinations; always a subspace.
4. **Independent**: only the trivial combination gives 0. **Spanning**: generates everything.
5. **Basis** = independent + spanning $\Rightarrow$ unique coordinates. **Dimension** = size of any basis.
6. **Injective**: no collisions. **Surjective**: everything hit. **Bijective**: both.
7. **Linear map**: $L(au + bv) = aL(u) + bL(v)$; $L(0) = 0$; determined by a basis.
8. **Kernel**: what's crushed to 0; injective $\iff$ $\ker = \{0\}$. **Range**: what's reachable; surjective $\iff$ range $= V$.
9. **Rank–nullity**: $\operatorname{rank} L + \dim \ker L = \dim U$. Kept + lost = total.
10. **ML**: data = vectors, features dependent = redundancy, Wx linear / $Wx+b$ affine, kernel = lost info, PCA = project on a k-dim subspace.

Next episode: matrices (Lecture 2). Stay tuned, bestie.